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Internship Report

The Kuramoto model with inertia on graphs and its applications to power systems

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Introduction

These notes report on an internship made in Spring 2022 at IMJ-PRG, under the supervision of David Gérard-Varet, Professor at Université Paris Cité, in order to obtain a Master's Degree in Mathematics and Modelling at Sorbonne Université. The topic of the internship was the system of differential equations known as the Kuramoto model with inertia, governing a network of oscillators interacting with each other. The main goal was to model a power grid and to understand the behaviour of large scale systems, with potential applications to the stability and control of power networks in mind. However the applications of the Kuramoto model are numerous, and the contents of these notes could prove useful in many situations where synchronisation phenomena happen.

I first contacted David Gérard-Varet in November of 2021 and we met at Université Paris Cité the following week. He introduced me to the Kuramoto Model, which I had never heard about, and explained me what he had in mind for a potential internship: comparing various models with inertia to study electrical power grids. This quickly resonated with me and I started working under his supervision in April after my Master courses ended. I first studied the series of papers by Medvedev et. al. [7, 8, 28, 35] on the mean field limit of the Kuramoto Model on graphs, and then transposed these techniques to our model. In parallel I got familiar with electrical engineering literature on large scale power systems [22, 38, 48]. I then turned to performing a rigorous mathematical treatment of the results in those latter papers. We exchanged on a biweekly basis with David Gérard-Varet and I once again thank him for his time and always insightful advice.

These notes are divided as follows: In the first chapter, we introduce the Kuramoto model and do a very brief overview of the dense literature on the subject. We will cover the original model proposed by Kuramoto in 1975 and its progressive generalizations, up to the case of our interest: the Kuramoto model with inertia on graphs.

Chapter 2 is dedicated to the modelling of a power network, deriving the Kuramoto model from the usual swing equation of power systems.

In the remaining chapters we conduct a theoretical analysis of the model, as the number of machines in the network goes to infinity. Chapter 3 covers the case of dense graphs, when every node of the network is connected through a power line to a large number of other nodes. Using mean field theory we will study the convergence of the Kuramoto system of equations to a single partial differential equation.

Finally, Chapter 4 deals with the case of nearest-neighbour graphs, where every node is only connected to its neighbours. In this setting we show that the Kuramoto model with inertia converges to a continuous wave equation under suitable hypothesis, a result that had been formally derived in electrical engineering works.

1 The Kuramoto model

1.1 The original Kuramoto model and its mean field limit

Synchronisation of oscillations is a widespread phenomena in non-linear science [40, 46], from biology [52] to social sciences [27, 36] and neuroscience [4, 33, 49]. In his pioneering work on chemical instabilities, Kuramoto introduced a model describing the dynamics of interacting oscillators [29, 30]. It turned out a very successful approach to the general problem of synchronization in a population of interacting elements. The original Kuramoto model is a system of n ordinary differential equations for the phases $(\theta_i)_{1\leq i\leq n}$ of the oscillators, writing as

$$\frac{d}{dt}\theta_i(t) = \omega_i + \frac{K}{n}\sum_{j=1}^{n}\sin(\theta_j(t) - \theta_i(t)).$$

The quantities $(\omega_i)_{1\leq i\leq n}$ model intrinsic frequencies of the oscillators and are usually sampled independently from a probability distribution with density $g(\omega)$. The prefactor K on the interaction term is the strength of the coupling. Extensive theoretical and numerical studies have followed Kuramoto's original work, providing detailed insight on the dynamics of the system (to cite a few, [1, 5, 12, 13, 31, 41, 44] and the references therein). The behaviour is vastly influenced by the value of the coupling strength K. For instance, asymptotic phase locking can be shown [26, 50] for K great enough: the limit

$$\lim_{t\to\infty} |\theta_i(t) - \theta_j(t)|$$

exists for every i, j, provided the initial phases are not too spread apart. On the other hand, K.A.M. theory shows that for small enough K the dynamics is conjugated to the system without interaction, that is $K = 0$, meaning there is little to no synchronization. We refer to [45] for a more in-depth take on those results.

In many cases of interest, the population of oscillators is extremely large and one has no hope to efficiently solve the system, even numerically. It is however possible to approximate the Kuramoto model for large n by a continuous model, and many of the insights and advances on the finite model do in fact stem from this limit object. This is the so called mean field approach for large particle systems. Mean field analysis is a wide topic, and we refer to [25] for a general course on the subject. Its application to the Kuramoto model [43] essentially follows the usual approach and thus illustrates quite well the general theory. The key idea is to describe the system at a given time using a probability measure instead of a collection of phases and intrinsic frequencies. This is done by introducing the empirical measure:

$$\mu_t^n = \frac{1}{n}\sum_{i=1}^{n}\delta_{(\theta_i(t),\omega_i)}.$$

As $n \to \infty$, we can reasonably expect that the measure is well approximated by an absolutely continuous measure μ_t with density $\rho(t, \theta, \omega)$. The density ρ should solve (at least in a weak sense) the following partial differential equation, called the mean field PDE:

$$\partial_t\rho(t, \theta, \omega) + \partial_\theta\big(\rho(t, \theta, \omega)V[\rho(t)](\theta, \omega)\big) = 0.$$

The velocity field created by a distribution of oscillators $\rho(t)$ is given by

$$V[\rho(t)](\theta,\omega) = \omega + K\int_{\mathbb{T}\times\mathbb{R}} \sin(\xi-\theta)\rho(t,\xi,\varpi)d\xi d\varpi.$$

The interpretation of this PDE is quite straightforward: the quantity $\rho(t,\theta,\omega)d\theta d\omega$ describes the proportion of oscillators with phase and frequency in the infinitesimal volume $[\theta,\theta+d\theta]\times[\omega,\omega+d\omega]$ of the phase space. These oscillators have intrinsic frequency ω, and are subject to a cumulated force due to the oscillators with phases and frequencies close to ξ and ϖ of magnitude $\sin(\xi-\theta)\rho(t,\xi,\varpi)d\xi d\varpi$, hence the form of the velocity field.

The mean field approximation is mathematically justified because of the following property [32] (which is common in mean field models [17, 25, 37]): If d_{BL} is the Bounded Lipschitz distance between two probability measures on $\mathbb{T}\times\mathbb{R}$, then there exists a constant C depending on a finite time interval $[0,T]$ such that for any time $t\in[0,T]$, the following inequality holds:

$$d_{BL}(\mu_t^n,\mu_t) \leq C d_{BL}(\mu_0^n,\mu_0).$$

For large n it is possible to approximate the initial density of the PDE: for instance, if the initial data of the finite model is sampled randomly with law μ_0, then by the Law of Large Numbers $d_{BL}(\mu_0^n,\mu_0)$ should be small, hence the system being correctly approximated by the solution of the mean field PDE.

Working with the continuum limit enables one to apply PDE techniques to obtain results on the asymptotic behaviour of the solutions. The degree of synchronisation of the system can be described using the order parameter

$$r(t)e^{i\psi(t)} = \int_{\mathbb{T}\times\mathbb{R}} e^{i\theta}\mu_t(d\theta,d\omega).$$

Small values of the modulus r correspond to incoherent behaviour of the oscillators: in particular the stationary uniformly distributed state

$$\rho(t,\theta,\omega) = \frac{1}{2\pi}g(\omega)$$

has order parameter $r=0$. On the other hand, $r=1$ would correspond to total synchronisation, with μ_t being concentrated on one phase $\theta(t)$. Note that the order parameter also allows us to rewrite the velocity field as:

$$V[\rho(t)](\theta,\omega) = \omega + Kr(t)\sin(\psi(t)-\theta).$$

For a given $r>0$, stationary solutions called partially locked states (PLS) [15] describe the partial synchronisation of the oscillators. Their expression is of the form:

$$\rho_{PLS}(\theta,\omega) = \begin{cases} g(\omega)\left(\alpha(\omega)\delta_{\arcsin(\frac{\omega}{Kr})} + (1-\alpha(\omega))\delta_{\pi-\arcsin(\frac{\omega}{Kr})}\right) & \text{for } |\omega|\leq Kr \\ \\ g(\omega)\dfrac{\sqrt{\omega^2-K^2r^2}}{2\pi\left|\omega - Kr\sin(\theta)\right|} & \text{for } |\omega|>Kr, \end{cases}$$

with an arbitrary function α distributing between the two equilibrium. The derivation of this expression comes from solving the stationary mean field PDE with the ansatz of a given r and $\psi=0$, and is a good exercise on distributions.

Fully rigorous mathematical proofs are rather recent [13, 14, 15, 21] and require some hypothesis on the distribution of frequencies g. In the simplest case of continuous unimodal symmetric g, the asymptotic behaviour is as follows [45]:

• For $K < K_c = \frac{2}{\pi g(0)}$, the uniform state $\rho_0 = \dfrac{1}{2\pi} g(\omega)$ is asymptotically stable, and $r(t) \to 0$. This is the analogue of Landau damping in the Vlasov equation [51].

• For $K = K_c$ this states becomes unstable.

• For greater $K > K_c$, if the initial data is sufficiently close to some PLS, up to a change of rotating frame, $\rho(t) \to \rho_{PLS}$ (in an appropriate topology), altogetjer with convergence of the order parameter $r(t) \to r_\infty$, with r_∞ corresponding to the PLS.

This overview of the model and its mean field limit is far from exhaustive and only begins to showcase both the complexity and the pertinence of Kuramoto's model. We refer to [16] for a more complete take on the recent works on the continuum limit, and now turn to the extension of the Kuramoto model on graphs.

1.2 The Kuramoto model on graphs with inertia

In its original formulation, the Kuramoto model has a unique global coupling strength K, and every oscillator interacts with each other. However, in many situations, it might be of interest to take into account the topology of the network of oscillators. Additionally, in many physical situations, such as in biology [19] and especially electrical engineering [22], it is pertinent to consider a second order derivative in time to add inertia to the system. This leads to a natural generalisation of the Kuramoto model,

$$\lambda \frac{d^2}{dt^2}\theta_i(t) + \frac{d}{dt}\theta_i(t) = \omega_i + \frac{K}{n}\sum_{j=1}^{n} W_{i,j}^n \sin(\theta_j(t) - \theta_i(t)),$$

where λ should be a non-negative coefficients if we expect the system to be stable. The coupling strength K has been replaced by the coefficients $(W_{i,j}^n)_{1\leq i,j\leq n}$ which model different interaction strengths between oscillators (which can be symmetric or not). Hence $(W_{i,j}^n)_{1\leq i,j\leq n}$ can be seen as the adjacency matrix of a (possibly directed) weighted graph on n vertices.

In the inertia-less case ($\lambda = 0$), series of works by Chiba, Medvedev et. al. [7, 8, 9, 28, 35] and Coppini et. al. [3, 11] have covered the mean field analysis of the Kuramoto model on graphs. Those works relies on the theory of graphons [34] to model graphs on arbitrary large number of vertices. The limit object of the sequence of graphs W^n is a function W on the unit square $[0,1]^2$, where $W(x,y)$ can be thought of as the edge weight between vertices around x and y in an infinite vertex set $[0,1]$. While we will not go into the details of the theory, we make a brief introduction to the topic in section 3.5 and refer to [24] for a more in-depth course. Several results from the all-to-all coupled case have been extended and adapted to the model on graphs. For instance, there exists critical values of the coupling strength K_c depending on the graphon W, for which for $K < K_c$ the uniform incoherent state is stable [7]. Although the inertia model on graphs has yet to receive as much mathematical coverage as the original one, Chiba and Medvedev have also recently studied the stability of the uniform state [10].

The first purpose of these notes is to provide a rigorous statement of the mean field analysis for the model with inertia, and the convergence of the finite system. It will be developed in Chapter 3 and the method will be mainly adapted from the first order case treated in [28].

Because the framework of graphons only applies to dense enough graphs, the mean field approach is only possible under hypothesis on the graph sequence as $n \to \infty$. Notably, the interaction strength between two individual oscillators must remain bounded and every oscillator must interact with a large number of others. Although the recent theory of L^p graphons and graphops allows for an analysis of sparser graphs [23, 35], it still does not cover the case of graphons with bounded average degree such as nearest-neighbour graphs. The second goal of these notes is thus to initiate a study of the Kuramoto model on some nearest neighbour graphs.

Some results have been formally derived in electrical engineering works [38, 48]. Considering a power network with n machines or generators located at positions denoted by $(x_i^n)_{1\leq i\leq n}$ that interact with their close neighbours through a power line. This corresponds to taking $W_{i,j}^n \neq 0$ if and only if machines i and j are neighbours (in a sense to be precisely defined, which is a generalisation of being adjacent on a Cartesian grid). One can then expect convergence as $n \to \infty$ of the Kuramoto model with inertia to a continuous wave equation

$$\lambda \partial_t^2 \theta(t,x) + \partial_t \theta(t,x) = \Delta \theta(t,x) + \omega(x).$$

Indeed, if we take for instance a simple 1-dimensional network $x_i^n = \frac{i}{n}$, and $W_{i,j}^n = 1$ if $|i-j| = \frac{1}{n}$, the interaction term writes

$$\sum_{j=1}^{n} W_{i,j}^n \sin(\theta_j(t) - \theta_i(t)) = \sin(\theta_{i+1}(t) - \theta_i(t)) + \sin(\theta_{i-1}(t) - \theta_i(t)).$$

This is a non-linear discretization of the second order derivative, and it should converge up to a correct rescaling, provided we can replace the sine by its first order Taylor approximation. This problem is very related to the convergence of numerical schemes and discrete approximations of differential operators, especially finite volume methods [20]. The exact framework and the proof of this convergence is the purpose of Chapter 4.

We end here the overview of the Kuramoto model and its generalisations. We advise any interested reader to consult the previously quoted works and the references therein. The next chapter is devoted to the modelling of a power grid which will lead to a Kuramoto model with inertia on a graph.

2 A model for power grids

The purpose of this chapter is to justify the use of the Kuramoto Model as a way to describe large-scale power systems, following the approach of [22].

We consider a geographical region of space Ω, such as a country or town, spanned by a power grid composed of n components, that are either loads (consuming power) or generators (producing power), connected through power lines. The i-th component of the network is located at the point $x_i \in \Omega$. The power grid is assumed to run on alternative current at a frequency ω_0, so that every component has instantaneous frequency close to ω_0. The goal is to derive a system of equation describing the deviation from the root frequency of the voltage phases of the circuit.

2.1 Mechanical balance

We model a machine or generator as a rotor with moment of inertia J. The angle of the rotor at time t can be written $\omega_0 t + \theta(t)$, where by hypothesis

$$\frac{d}{dt}\theta(t) \ll \omega_0.$$

Machines produces mechanical or heat energy using electrical power while generators use such energy to produce electricity. We denote by P the power *produced* by the component, so that $P^{ext} > 0$ for machines and $P^{ext} < 0$ for generators. This power is converted into either electrical power, kinetic energy or dissipated through friction, leading to the following power balance:

$$\frac{J}{2}\frac{d}{dt}\left(\omega_0 + \frac{d}{dt}\theta(t)\right)^2 + \alpha\left(\omega_0 + \frac{d}{dt}\theta(t)\right)^2 + P^{elec} + P^{ext} = 0,$$

with $\alpha > 0$ the friction coefficient and P^{elec} the electrical power *generated* by the component (so $P^{elec} < 0$ for machines, $P^{elec} > 0$ for generators). Expanding this expression, we find, in dimensionless form:

$$\left(1 + \frac{1}{\omega_0}\frac{d}{dt}\theta(t)\right)\frac{J}{\alpha\omega_0}\frac{d^2}{dt^2}\theta(t) + \left(2 + \frac{1}{\omega_0}\frac{d}{dt}\theta(t)\right)\frac{1}{\omega_0}\frac{d}{dt}\theta(t) + 1 + \frac{P^{elec} + P^{ext}}{\alpha\omega_0^2} = 0.$$

As in [22], we suppose the rate at which the rotor stores kinetic energy is far less than the rate at which it dissipates energy, that is:

$$\frac{J}{\alpha\omega_0}\frac{d^2}{dt^2}\theta(t) \ll 1,$$

a common condition in mechanical systems. Keeping only the terms up to first order in the former equation yields:

$$\frac{J}{\alpha\omega_0}\frac{d^2}{dt^2}\theta(t) + \frac{2}{\omega_0}\frac{d}{dt}\theta(t) + 1 + \frac{P^{elec} + P^{ext}}{\alpha\omega_0^2} = 0. \tag{1}$$

The power P^{ext} is a given quantity, but it remains to express P^{elec} using the network topology to close the system.

2.2 Electrical balance

We now add indices to the powers, phases and other quantities to precise which component of the grid we are talking about. By Faraday's law, the rotation of the rotor generates an electromotive force oscillating at the same frequency. We write in complex formalism the voltage and intensity phasors of component k (measured in generator convention):

$$V_k(t) = \mathcal{V}_k e^{i(\omega_0 t + \theta_k(t))}$$

$$I_k(t) = \mathcal{I}_k e^{i(\omega_0 t + \psi_k(t))}$$

As before we have

$$\frac{d}{dt}\psi_k(t), \frac{d}{dt}\theta_k(t) \ll \omega_0.$$

The average electrical power generated by component k at time t is the average over a period:

$$P_k^{elec} = \frac{\omega_0}{2\pi} \int_{t-\frac{2\pi}{\omega_0}}^{t} \mathrm{Re}(V_k(s))\mathrm{Re}(I_k(s))ds.$$

By hypothesis, we can approximate $\theta_k(s)$ by $\theta_k(t)$ on the period $[t - \frac{2\pi}{\omega_0}, t]$. A direct computation then gives:

$$P_k^{elec} = \frac{1}{2}\cos(\theta_k(t) - \psi_k(t)) = \frac{1}{2}\mathrm{Re}(V_k(t)\bar{I}_k(t)).$$

We let $a_{kj} = 1$ if components j and k are connected through a power line, and 0 otherwise. Using Kirchhoff's Law, one can express $I_k(t)$ as

$$I_k(t) = \sum_{j=1}^{n} a_{kj} I_{kj}(t),$$

where I_{kj} is the intensity of the current flowing from k to j. This intensity can be further expanded using generalized Ohm's Law for phasors:

$$I_{kj}(t) = Z_{kj}^{-1}(V_k(t) - V_j(t)),$$

where $Z_{kj} = R_{kj} + iX_{kj}$ is the complex impedance of the power line. Plugging those expressions in the formula for P_k^{elec} yields

$$\begin{aligned}
P_k^{elec} &= \frac{1}{2}\sum_{j=1}^{n} a_{jk}\mathrm{Re}\left(V_k(t)\bar{Z}_{kj}^{-1}\left(\bar{V}_k(t) - \bar{V}_j(t)\right)\right) \\
&= \frac{1}{2}\sum_{j=1}^{n} a_{jk}\left|Z_{kj}\right|^{-2}\mathrm{Re}\left(Z_{kj}\left(\mathcal{V}_k^2 - \mathcal{V}_k\mathcal{V}_j e^{i(\theta_k(t)-\theta_j(t))}\right)\right) \\
&= \frac{1}{2}\sum_{j=1}^{n} a_{jk}\left|Z_{kj}\right|^{-2} R_{kj}\mathcal{V}_k^2 \quad - \quad \frac{1}{2}\sum_{j=1}^{n} a_{jk}\left|Z_{kj}\right|^{-2}\mathcal{V}_k\mathcal{V}_j\mathrm{Re}\left(Z_{kj}e^{i(\theta_k(t)-\theta_j(t))}\right)
\end{aligned}$$

The real part in the second sum can be made explicit using polar coordinates:

$$Z_{kj} = \left|Z_{kj}\right| e^{i\varphi_{kj}}, \quad \varphi_{kj} = \arctan\left(\frac{X_{kj}}{R_{kj}}\right),$$

leading to

$$\begin{aligned}
\mathrm{Re}\left(Z_{kj}e^{i(\theta_k(t)-\theta_j(t))}\right) &= \left|Z_{kj}\right|\cos\left(\varphi_{kj} + \theta_k(t) - \theta_j(t)\right) \\
&= \left|Z_{kj}\right|\sin\left(\frac{\pi}{2} - \varphi_{kj} - \theta_k(t) + \theta_j(t)\right)
\end{aligned}$$

The expression of P_k^{elec} then reduces to:

$$P_k^{elec} = P_k - \frac{1}{2}\sum_{j=1}^{n} a_{jk}\left|Z_{kj}\right|^{-1}\mathcal{V}_k\mathcal{V}_j\sin\left(\theta_j(t) - \theta_k(t) - \beta_{kj}\right) \tag{2}$$

where $\beta_{kj} = \varphi_{kj} - \frac{\pi}{2}$, and P_k is a constant power given by

$$P_k = \frac{1}{2} \sum_{j=1}^{n} a_{jk} \left| Z_{kj} \right|^{-2} R_{kj} \mathcal{V}_k^2.$$

In the sequel, we will assume that β_{kj} is equal to a constant β. This is the case if we assume that the impedance Z_{kj} is given by an impedance per unit of length $z = r + ix$ through the formula

$$Z_{kj} = z \left| x_j - x_k \right|,$$

because then the ratio $\dfrac{X_{kj}}{R_{kj}} = \dfrac{x}{r}$ is constant.

2.3 Closing the model

We can now write a closed system of equations by combining the n mechanical power balances and the expression of P_k^{elec}. We suppose that all components of the grid have the same moment of inertia J and friction coefficient α.

The mechanical balance (1) for the k-th component writes:

$$\frac{d^2}{dt^2}\theta_k(t) + \frac{2\alpha}{J}\frac{d}{dt}\theta_k(t) = -\frac{\alpha\omega_0}{J} - \frac{P_k^{ext}}{J\omega_0} - \frac{P_k^{elec}}{J\omega_0}.$$

Plugging in the expression of P_k^{elec} from (2), we discover:

$$\frac{d^2}{dt^2}\theta_k(t) + \frac{2\alpha}{J}\frac{d}{dt}\theta_k(t) = \omega_k + \frac{1}{2J\omega_0} \sum_{j=1}^{n} a_{jk} \left| Z_{kj} \right|^{-1} \mathcal{V}_k \mathcal{V}_j \sin\left(\theta_j(t) - \theta_k(t) - \beta\right),$$

where

$$\omega_k = -\frac{\alpha\omega_0}{J} - \frac{P_k^{ext}}{J\omega_0} - \frac{P_k}{J\omega_0}.$$

Defining the interaction coefficients

$$W_{kj} = \frac{a_{jk}\mathcal{V}_k\mathcal{V}_j}{2J\omega_0 \left| Z_{kj} \right|},$$

and the constant $D = \frac{2\alpha}{J}$, we obtain the Kuramoto model with a phase shift β:

$$\frac{d^2}{dt^2}\theta_k(t) + D\frac{d}{dt}\theta_k(t) = \omega_k + \sum_{j=1}^{n} W_{kj} \sin\left(\theta_j(t) - \theta_k(t) - \beta\right).$$

This Kuramoto model with inertia on a graph, referred to in the sequel as Kuramoto model, will be the main object of the theoretical study that follows. The analysis is divided in two independent paradigms: first will be treated the case of dense networks, for which the usual mean field approach will be conducted. In a second part, we will study the convergence on nearest neighbour graphs, as done in [38, 48]. These graphs are further away from the usual framework of most results on the Kuramoto model, but seem to be more accurate models of real life power grids [39, 42] which are rather sparse.

Remark 2.3.1. *The hypothesis of all moments of inertia and friction coefficients being the same is used to fix the constant D, which would otherwise be a D_k depending on the component of the grid. Although this case is not considered in the sequel, we believe under suitable assumptions (similar to the ones that will be made on the ω_k) the mean field analysis can still be conducted.*

3 Mean field analysis on dense graphs

3.1 Setup and notations

This chapter covers the convergence of the second order Kuramoto model on dense graphs. Broadly, a dense graph is a graph for which the number of edges (i.e. the number of $W_{ij}^n \neq 0$ in our model) is close to the maximum possible number of edges n^2. In this case, we expect to be able to carry out the mean field analysis, as in the original all-to-all coupled oscillators case (which corresponds to $W_{ij}^n \equiv 1$). The analysis in this chapter is mainly adapted from [28] where the model without inertia was treated.

We now set some notation. As in Chapter 2, consider a bounded set $\Omega \subset \mathbb{R}^d$ of Lebesgue measure $|\Omega| = 1$, where the power grid lies. For each $n \in \mathbb{N}^*$, we consider $(x_i^n)_{1\leq i\leq n}$ a sequence of points in Ω that represent the positions of n generators or machines. We also fix a partition $(P_i^n)_{1\leq i\leq n}$ of Ω such that $x_i^n \in P_i^n$ and $|P_i^n| > 0$, used to construct step functions on Ω. We consider a graph on the vertex set $(x_i^n)_{1\leq i\leq n}$ and with an adjacency matrix $(W_{ij}^n)_{1\leq i,j\leq n}$ to model the power lines.

As derived in the previous section, we consider the following Kuramoto model with inertia for n oscillators $(\theta_i^n)_{1\leq i\leq n}$ at the edges of this graph, given by:

$$\text{(KMn)}: \quad \frac{d^2}{dt^2}\theta_i^n(t) + D\frac{d}{dt}\theta_i^n(t) = \omega_i^n + \frac{1}{n}\sum_{j=1}^{n} W_{ij}^n \sin(\theta_j^n(t) - \theta_i^n(t) - \beta)$$

with initial conditions denoted by

$$\theta_i^n(0) = a_i^n \in \mathbb{T}, \ \frac{d}{dt}\theta_i^n(0) = b_i^n \in \mathbb{R}. \tag{3}$$

Using the partition (P_i^n) to build piecewise constant functions, we can without loss of generality suppose that the ω_i^n are given by a function ω^n through the formula:

$$\omega^n(x) = \omega_i^n \text{ for (almost all) } x \in P_i^n.$$

We could also draw the intrinsic frequencies randomly to account for imperfections of the machines. The analysis is basically the same, and we briefly sketch the differences in section 3.4.

Likewise, we consider that W_{ij}^n is given by a function W^n which is the *graphon* of the graph:

$$W^n(x, y) = W_{ij}^n \text{ for } x \in P_i^n, \ y \in P_j^n$$

Graphons provide a natural framework to study dense graph limits. The hypothesis of dense graphs translates in terms of graphons to the assumption that the graphons W^n do not converge to the zero function. This can only happen for graphs with enough edges so that the

support of W^n does not vanish, hence the name *dense graph*. Although we will only use basic properties of graphons, we refer an interested reader to section 3.5 and to [24] for a detailed overview of the topic.

Following the all-to-all case, our results will focus on the family of empirical measures, each describing the distribution of phases and instantaneous velocity (θ, ϕ) at a point x and a time t:

$$(\mu^n)_t^x = \delta_{(\theta_i^n(t), \frac{d}{dt}\theta_i^n(t))} \text{ for } x \in P_i^n.$$

Under suitable assumptions, we show that the behaviour of the empirical measures is approximated for large n by the following mean field equation for the distribution of phases and velocities $\mu(t, x, d\theta, d\phi)$:

$$(\text{MF}) : \partial_t \mu(t,x) + \phi\ \partial_\theta \mu(t,x) + \partial_\phi\big(\mu(t,x)\ V[\mu(t)](x,\theta,\phi)\big) = 0$$

with the acceleration field V given by:

$$V[\mu(t)](x,\theta,\phi) = \omega(x) - D\phi + \int_\Omega W(x,y) \int_{\mathbb{T}\times\mathbb{R}} \sin(\xi - \theta - \beta)\mu(t,y,d\xi,d\psi)dy$$

where the functions ω and W are the limits of ω^n and W^n.

For the sake of conciseness, in the sequel we will sometimes only refer to the *distribution of phases*, but it should always be understood as the *distribution of phases* ***and*** *their instantaneous velocity* $\Theta = (\theta, \phi)$ and not only θ. Finally, in all the proofs, C will denote a constant that may change from line to line, and, if relevant, its dependence is precised in the statement of the results.

We end this first section with two remarks on the physical meaning of Ω, W and ω ; and on the definition of empirical measure:

Remark 3.1.1. *In our problem, the graph has a geographical reality, being a power grid. We thus let Ω be an arbitrary region of space, which can be useful to build realistic graphs: for instance, we might be given a function $W(x,y)$ giving the density of power connections between two locations, from which we can build accurate coefficients W_{ij}^n. If the spatial distribution of the oscillators is not of interest, one can simply fix $\Omega = [0,1[$ and $P_i^n = [\frac{i-1}{n}, \frac{i}{n}[$, as it is usually done both for the Kuramoto model on graphs [7, 11] and the general graphon theory [24, 34].*

Remark 3.1.2. *In the all-to-all coupled model, the empirical measure is defined as*

$$(\mu^n)_t = \frac{1}{n}\sum_{i=1}^n \delta_{(\theta_i^n(t), \frac{d}{dt}\theta_i^n(t))}.$$

For the model on graphs, this definition does not encompass all information any more, since we need to keep track of the vertices the oscillators are located at. A straightforward generalization would be to consider probability measures on the space $\mathbb{T}\times\mathbb{R}\times\Omega$, and let:

$$(\mu^n)_t = \frac{1}{n}\sum_{i=1}^n \delta_{(\theta_i^n(t), \frac{d}{dt}\theta_i^n(t), x_i^n)}$$

There is an issue with this choice, which is a bit technical: since we use the Bounded Lipschitz distance on the space of probability measures, the analysis would require the limit graphon W

to be a Lipschitz function, which is quite a restriction. The way to overcome this is to observe that the x variable does not play a role other than a parameter in both (KMn) and (MF). In particular, in both equations the marginals in x of the measures do not evolve over time. If the x_i^n are uniformly distributed in Ω, we can expect the empirical marginal

$$\sum_{i=1}^{n} \delta_{x_i^n}$$

to converge to the Lebesgue measure on Ω, which would be the marginal of the mean field limit as well. The trick is to fix the marginal in x for all measures in play, and thus define the empirical measure in a way such that it already has as a marginal the Lebesgue measure, as we did. Since then all the distributions have the same marginal, we can use more precise distances to compare them, which do not require W to be Lipschitz. To better understand the differences, one can compare the analysis done in the following and in [28], to the analysis in [7] where the Lipschitz hypothesis on W is made.

The chapter is divided as follows: In section 3.2, we introduce the metric spaces for the empirical measures, in which the analysis will be carried out. In the following section 3.3, we will derive a fixed point equation for the distribution of phases, that is satisfied by the empirical measures of the Kuramoto Model. We will solve this equation and show continuous dependence of the solution with respect to the initial data, the graphon and the intrinsic frequencies. Section 3.4 introduces the notion of weak (measure) solution of the mean field limit, and convergence of the empirical measures to the mean field limit is shown with deterministic and random initial data. We also briefly show how to adapt the previous analysis in the case of randomly drawn intrinsic frequencies. In section 3.5 we briefly motivate the elements of graphon theory that we use.

3.2 The metric spaces

The empirical measures are sums of Dirac δ measure since given exact initial data, the phase at time t of any oscillator is entirely determined. However, we could imagine that the initial data is random or uncertain, or that there are several oscillators in the neighbourhood P_i^n of x, whose phases are initially distributed according to a certain measure μ_0^x. Then, at a given time t, the oscillators at x would not be correctly described by a single phase but by a distribution of phases μ_t^x. Since the system (KMn) can only apply to a finite number of oscillators, we have to seek a more general equation for μ_t^x to model such a situation.

Our first step is to rigorously define the metric space of the distribution of phases. We denote by $\mathcal{M} = \mathcal{M}(\mathbb{T} \times \mathbb{R})$ the space of probability measures on $\mathbb{T} \times \mathbb{R}$. We will use the following Bounded Lipschitz distance between two probability measure μ, ν:

$$d(\mu, \nu) = \sup_{f \in BL} \left| \int f(d\mu - d\nu) \right|.$$

Here $BL = BL(\mathbb{T} \times \mathbb{R})$ is the set of Lipschitz continuous functions from $\mathbb{T} \times \mathbb{R}$ to $[0, 1]$ with Lipschitz constant at most 1. It can be shown that d metricizes the topology of weak convergence of measures on $\mathcal{M}$ and that the space is complete ([18], Theorem 11.3.3). At a

fixed time t, the system is described by the collection of probability measures describing the phase distribution at each point, $(\mu^x)_{x\in\Omega}$. We thus introduce the following space: Consider

$$\bar{M} = \{f : \Omega \to \mathcal{M} \text{ measurable}\}$$

where f measurable means that $x \mapsto f(x)(B)$ is measurable for any Borel subset $B \subset \mathbb{T}\times\mathbb{R}$. If $\sim$ denotes the equivalence relation of dx-almost everywhere equality with dx the Lebesgue measure on Ω, then we define the quotient space

$$\bar{\mathcal{M}} = \bar{M}/_\sim$$

We endow $\bar{\mathcal{M}}$ with the following distance: for $\bar{\mu} \in \bar{\mathcal{M}}$, we use the notation μ^x for $\bar{\mu}(x)$ (which is an element of $\mathcal{M}$). Then $\bar{d}$ is defined as:

$$\bar{d}(\bar{\mu}, \bar{\nu}) = \int_\Omega d(\mu^x, \nu^x)dx.$$

Note that this integral is always finite because $d(\mu^x, \nu^x) \leq 2$. The metric space $(\bar{\mathcal{M}}, \bar{d})$ is complete (the proof is the same as for the completeness of the standard Lebesgue space $L^1(\Omega)$, see [28]). Finally, to describe the evolution of the system over time, we define for $T > 0$ the space of continuous functions with values in $\bar{\mathcal{M}}$:

$$\dot{\mathcal{M}}_T = \mathcal{C}([0, T], \bar{\mathcal{M}}).$$

For any parameter $\alpha \in \mathbb{R}$, $(\dot{\mathcal{M}}_T, d_\alpha)$ is also complete, where

$$d_\alpha(\dot{\mu}, \dot{\nu}) = \sup_{t\in[0,T]} e^{-\alpha t}\bar{d}(\bar{\mu}_t, \bar{\nu}_t),$$

with the notation $\bar{\mu}_t = \dot{\mu}(t)$ for $\dot{\mu} \in \dot{\mathcal{M}}_T$.

In the sequel, we will always denote with dots the elements $\dot{\mu}$ of $\dot{\mathcal{M}}_T$, with bars the elements $\bar{\mu}$ of $\bar{\mathcal{M}}$, and use no ornament for measures $\mu \in \mathcal{M}$.

3.3 The fixed point equation

3.3.1 From the equation of characteristics to the fixed point equation

Our next goal is to write a fixed point equation in $\dot{\mathcal{M}}_T$ that would be a natural generalisation of (KMn) to any distribution of phases. Indeed, (KMn) only applies to a finite number of oscillators (described by a sum of Dirac masses) with piecewise constant functions W and ω, while the mean field limit is adapted for distribution of phases with density, and general W and ω. The fixed point equation we will derive will give a wider (but more technical) point of view encompassing general measures, graphons and frequencies. We thus drop for a moment the dependence on n and fix a graphon $W \in L^\infty$ and a function $\omega \in L^1(\Omega)$. Suppose that we know beforehand the distribution of phases over time $\dot{\mu} \in \dot{\mathcal{M}}_T$. Then, letting $\Theta = (\theta, \phi)$, one can determine the evolution of an initial Θ_0 subject to the interaction with the oscillators given by $\dot{\mu}$ through the following equation of characteristics:

$$(\text{EC}) : \frac{d}{dt}\Theta(t) = \mathcal{V}^{\dot{\mu}}(t, x, \Theta(t))$$

with the velocity field being given by:

$$\mathcal{V}^{\dot\mu}(t,x,\Theta) = \begin{pmatrix} \phi \\ \omega(x) - D\phi + \int_\Omega W(x,y) \int_{\mathbb{T}\times\mathbb{R}} \sin(\xi-\theta-\beta)\mu_t^y(d\xi,d\psi)dy \end{pmatrix}.$$

Remark 3.3.1. *The ODE (EC) is the equation of characteristics for the mean field equation (MF). Indeed, (MF) can be explicitly rewritten as a continuity equation with velocity field $\mathcal{V}$:*

$$(MF): \partial_t \mu_t^x + \mathit{div}_\Theta\big(\mu_t^x \mathcal{V}^{\dot\mu}(t,x,\Theta)\big) = 0.$$

One can also notice that if $\dot\mu = \dot\mu^n$ is the empirical measure associated to the solution of (KMn) with initial conditions $(a_i^n, b_i^n)_{1\le i\le n}$, then (EC) with initial condition (a_i^n, b_i^n) gives the evolution of $(\theta_i^n, \frac{d}{dt}\theta_i^n)$ (provided we define correctly W and ω in terms of W_{ij}^n, ω_i^n).

We quickly remind Gronwall's lemma, that will be useful in many of the following proofs.

Lemma 3.3.2. *Let a, b be continuous real valued functions on $[0,T]$ such that*

$$a(t) \le C\int_0^t a(s)ds + \int_0^t b(s)ds + A$$

with $C>0$. Then the following inequality holds for all $t\in[0,T]$:

$$a(t) \le Ae^{Ct} + e^{Ct}\int_0^t e^{-Cs}b(s)ds.$$

The goal is to solve (EC) to generate a flow on the phase space $\mathbb{T}\times\mathbb{R}$, that will then dictate the evolution of the distribution of phases:

Lemma 3.3.3. *For any initial condition $\Theta_0 \in \mathbb{T}\times\mathbb{R}$, for almost every $x\in\Omega$, there exists a unique solution to (EC) on $[0,T]$. Equation (EC) generates for almost all x a flow $T_{t,s}^{\dot\mu}(x,\Theta)$ on $\mathbb{T}\times\mathbb{R}$ such that $T_{t,s}^{\dot\mu}(x,\Theta)$ is the solution of (EC) at time t which is equal to Θ at time s. It satisfies $T_{t,s}^{\dot\mu}(x,\cdot)^{-1} = T_{s,t}^{\dot\mu}(x,\cdot)$, is $\mathcal{C}^1$ in t, $\mathcal{C}^1$ in s, and globally Lipschitz continuous in Θ.*

Proof. In order to apply the Cauchy-Lipschitz theorem, it suffices to show that $\mathcal{V}^{\dot\mu}$ is continuous in time and globally Lipschitz continuous in Θ. On one hand, we have

$$\begin{aligned}
\left|\mathcal{V}^{\dot\mu}(t,x,\Theta) - \mathcal{V}^{\dot\mu}(s,x,\Theta)\right| &= \left|\int_\Omega W(x,y)\int_{\mathbb{T}\times\mathbb{R}} \sin(\xi-\theta-\beta)\left(\mu_t^y(d\xi,d\psi) - \mu_s^y(d\xi,d\psi)\right)dy\right| \\
&\lesssim \int_\Omega |W(x,y)|\, d(\mu_t^y,\mu_s^y)dy \\
&\le \|W\|_{L^\infty}\, \bar d(\bar\mu_t,\bar\mu_s),
\end{aligned}$$

where we used the fact that sin is bounded and Lipschitz for the first inequality. On the other hand,

$$\begin{aligned}
&|\mathcal{V}^{\dot\mu}(t,x,\Theta) - \mathcal{V}^{\dot\mu}(t,x,\Theta')| \\
&\quad= \left|\begin{pmatrix} \phi-\phi' \\ -D(\phi-\phi') + \int_\Omega W(x,y)\int_{\mathbb{T}\times\mathbb{R}}(\sin(\xi-\theta-\beta)-\sin(\xi-\theta'-\beta))\mu_t^y(d\xi,d\psi)dy \end{pmatrix}\right| \\
&\quad\lesssim \left|\phi-\phi'\right| + \int_\Omega |W(x,y)| \int_{\mathbb{T}\times\mathbb{R}} \left|\sin(\xi-\theta-\beta)-\sin(\xi-\theta'-\beta)\right| \mu_t^y(d\xi,d\psi)dy \\
&\quad\lesssim \left|\phi-\phi'\right| + \int_\Omega |W(x,y)| \int_{\mathbb{T}\times\mathbb{R}} \left|\theta-\theta'\right| \mu_t^y(d\xi,d\psi)dy \\
&\quad\lesssim (1+\|W\|_{L^\infty})\left|\Theta-\Theta'\right|,
\end{aligned}$$

where the implicit constants depend on D but not $\|W\|_{L^\infty}$. The properties of the flow $T^{\dot\mu}_{t,s}(x,\cdot)$ then follow from the standard results on regularity of solutions to ordinary differential equation. We show the Lipschitz regularity, as the explicit constant will be useful later:

$$\begin{aligned}\frac{d}{dt}\left|T^{\dot\mu}_{t,0}(x,\Theta)-T^{\dot\mu}_{t,0}(x,\Theta')\right|^2 &= \left(\mathcal{V}^{\dot\mu}(t,x,T^{\dot\mu}_{t,0}(x,\Theta))-\mathcal{V}^{\dot\mu}(t,x,T^{\dot\mu}_{t,0}(x,\Theta'))\right)\left(T^{\dot\mu}_{t,0}(x,\Theta)-T^{\dot\mu}_{t,0}(x,\Theta')\right)\\ &\leq C(1+\|W\|_{L^\infty})\left|T^{\dot\mu}_{t,0}(x,\Theta)-T^{\dot\mu}_{t,0}(x,\Theta')\right|^2.\end{aligned}$$

Integrating in time from 0 to t and applying Gronwall's Lemma 3.3.2, we obtain:

$$\left|T^{\dot\mu}_{t,0}(x,\Theta)-T^{\dot\mu}_{t,0}(x,\Theta')\right| \leq \sqrt{e^{TC(1+\|W\|_{L^\infty})}}\left|\Theta-\Theta'\right|.$$

□

Now we can use the flow $T^{\dot\mu}_{t,0}(x,\cdot)$ to relate the system at time t and at initial time 0. Indeed, if μ^x_t is the distribution of phases at time t at point x, then it should respect the following relation:

$$\mu^x_t(T^{\dot\mu}_{t,0}(x,B)) = \mu^x_0(B)$$

for any Borel set $B\subset\mathbb{T}\times\mathbb{R}$. This formula simply states that the proportion of oscillators whose phase at time t is in $T^{\dot\mu}_{t,0}(x,B)$ is exactly the proportion that had initially its phase in B, since the phases evolve according to the flow $T^{\dot\mu}$. We can rewrite this identity as the fixed point equation on $\dot\mu$:

$$\text{(FP)}: \ \textit{For a.e. } x, \quad \forall t\in[0,T], \quad \mu^x_t = T^{\dot\mu}_{t,0}(x,\cdot)\sharp\mu^x_0$$

where $\sharp$ is the pushforward of measures. The empirical measures are then naturally solutions to this equation:

Lemma 3.3.4. *For any $n\in\mathbb{N}^*$, consider a $\mathcal{C}^2$ function*

$$[0,T]\ni t\mapsto(\theta^n_1,...,\theta^n_n)\in\mathbb{R}^n,$$

and its associated empirical measure $\dot\mu^n$:

$$(\mu^n)^x_t = \delta_{(\theta^n_i(t),\frac{d}{dt}\theta^n_i(t))} \textit{ for } x\in P^n_i.$$

Denote $a^n_i=\theta^n_i(0)$, $b^n_i=\frac{d}{dt}\theta^n_i(0)$ its initial conditions. Then $(\theta^n_1,...,\theta^n_n)$ is a solution of (KMn) with initial conditions a^n_i,b^n_i and parameters ω_i, W_{ij} if and only if $\dot\mu^n$ solves (FP) with the functions ω, W defined as:

$$\omega(x)=\omega_i \textit{ for } x\in P^n_i$$

$$W(x,y)=\frac{W_{ij}}{n|P^n_j|} \textit{ for } x\in P^n_i,\ y\in P^n_j$$

and with initial data:

$$(\mu^n)^x_0 = \delta_{(a^n_i,b^n_i)} \textit{ for } x\in P^n_i.$$

Proof. The proof illustrates Remark 3.3.1: Suppose $(\theta_1^n, ..., \theta_n^n)$ is solution of (KMn), then we can rewrite the equation for $\theta_i^n(t)$ as

$$\frac{d}{dt}\Theta_i^n(t) = \mathcal{V}^{\dot{\mu}^n}(t, x, \Theta(t))$$

for any $x \in P_i^n$, which in turn gives

$$\Theta_i^n(t) = T_{t,0}^{\dot{\mu}^n}(x, a_i^n, b_i^n)$$

for any $x \in P_i^n$. Then, since

$$T_{t,0}^{\dot{\mu}^n}(x, \cdot)\sharp(\mu^n)_0^x = \delta_{T_{t,0}^{\dot{\mu}^n}(x, a_i^n, b_i^n)} \text{ for } x \in P_i^n.$$

we conclude that $\dot{\mu}^n$ satisfies (FP). The converse implication is analogous. □

Remark 3.3.5. *We have to define $W(x, y)$ as $\dfrac{W_{ij}}{n|P_j^n|}$ and not simply W_{ij} as we did in the beginning of this chapter. This is a technical issue due to the fact that $|P_j^n| \neq \frac{1}{n}$ a priori, so without this correction the integral in the velocity field $\mathcal{V}^{\dot{\mu}}$ would yield the interaction sum in (KMn) but with $\frac{1}{n}$ replaced by $|P_j^n|$.*

The previous lemma shows that (FP) is the right formulation of the Kuramoto model in the space $\dot{\mathcal{M}}_T$.

3.3.2 Solving the fixed point equation

We will now show that (FP) has a solution in $\dot{\mathcal{M}}_T$ for any initial data $\bar{\rho} \in \bar{\mathcal{M}}$, using Banach's fixed point theorem for contraction mappings. Fix such an initial data, as well as functions ω and W and define the following map:

$$A = A[\bar{\rho}, \omega, W] : \begin{array}{ccc} \dot{\mathcal{M}}_T & \longrightarrow & \dot{\mathcal{M}}_T \\ \dot{\mu} & \longmapsto & (T_{t,0}^{\dot{\mu}}(x, \cdot)\sharp\rho^x)_{x\in\Omega, t\in[0,T]} \end{array}$$

To keep the proofs easy to read, we will not write explicitly the dependence in ρ, ω, W until we start considering different values for this parameters. The result of this subsection is the following:

Lemma 3.3.6. *The map A is well defined. Moreover, there exists $C_1 > 0$ (depending only on d, D) such that for any $\alpha > C_1(1 + \|W\|_{L^\infty})$, the map A is a contraction on $(\dot{\mathcal{M}}_T, d_\alpha)$, verifying*

$$d_\alpha(A\dot{\mu}, A\dot{\nu}) \leq \frac{1}{2} d_\alpha(\dot{\mu}, \dot{\nu})$$

The equation (FP) admits a unique solution in the space $\dot{\mathcal{M}}_T$ for any T.

Proof. Let $\dot{\mu} \in \dot{\mathcal{M}}_T$. To show that A is well defined, we need to show $A\dot{\mu}$ is continuous in time. Let $t, s \in [0, T]$, we have, using the definition of d and $\bar{d}$

$$\begin{aligned} \bar{d}(\overline{(A\dot{\mu})}_t, \overline{(A\dot{\mu})}_s) &= \int_\Omega d\big((A\dot{\mu})_t^x, (A\dot{\mu})_s^x\big) dx \\ &= \int_\Omega \sup_{f\in BL} \left| \int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\big(T_{t,0}^{\dot{\mu}}(x, \cdot)\sharp\rho^x(d\Theta) - T_{s,0}^{\dot{\mu}}(x, \cdot)\sharp\rho^x(d\Theta)\big)\right| dx. \end{aligned}$$

Using the formula $\int f(\Theta)\, T\sharp\rho(d\Theta) = \int f \circ T(\Theta)\rho(d\Theta)$ for pushforwards, we get:

$$\begin{aligned}\bar{d}(\overline{(A\dot{\mu})}_t - \overline{(A\dot{\mu})}_s) &= \int_\Omega \sup_{f\in BL} \left| \int_{\mathbb{T}\times\mathbb{R}} \big(f(T^{\dot{\mu}}_{t,0}(x,\Theta)) - f(T^{\dot{\mu}}_{s,0}(x,\Theta))\big)\rho^x(d\Theta)\right| dx \\ &\leq \int_\Omega \int_{\mathbb{T}\times\mathbb{R}} \min\left(2, \left|T^{\dot{\mu}}_{t,0}(x,\Theta) - T^{\dot{\mu}}_{s,0}(x,\Theta)\right|\right)\rho^x(d\Theta)dx,\end{aligned}$$

where we used that f is Lipschitz and bounded by 1. By continuity of the flow

$$\left|T^{\dot{\mu}}_{t,0}(x,\Theta) - T^{\dot{\mu}}_{s,0}(x,\Theta)\right| \to 0$$

as $|t-s| \to 0$ and we can apply Lebesgue dominated convergence theorem to get the result. This shows that A is well defined. Now take $\dot{\nu} \in \dot{\mathcal{M}}_T$ and let us compare $A\dot{\nu}$ and $A\dot{\mu}$. Fixing a t, we have:

$$\bar{d}(\overline{(A\dot{\mu})}_t - \overline{(A\dot{\nu})}_t) \leq \int_\Omega \int_{\mathbb{T}\times\mathbb{R}} \left|T^{\dot{\mu}}_{t,0}(x,\Theta) - T^{\dot{\nu}}_{t,0}(x,\Theta)\right| \rho^x(d\Theta)dx,$$

by repeating the exact same steps as before, and dropping the min. Call $a(t)$ the right-hand side. Its integrand verifies

$$T^{\dot{\mu}}_{t,0}(x,\Theta) - T^{\dot{\nu}}_{t,0}(x,\Theta) = \int_0^t \mathcal{V}^{\dot{\mu}}\big(s, x, T^{\dot{\mu}}_{s,0}(x,\Theta)\big) - \mathcal{V}^{\dot{\nu}}\big(s, x, T^{\dot{\nu}}_{s,0}(x,\Theta)\big)\, ds$$

since at initial time $T^{\dot{\mu}}_{0,0}(x,\Theta) = \Theta = T^{\dot{\nu}}_{0,0}(x,\Theta)$. We further write it as:

$$\begin{aligned}T^{\dot{\mu}}_{t,0}(x,\Theta) - T^{\dot{\nu}}_{t,0}(x,\Theta) = &\int_0^t \mathcal{V}^{\dot{\mu}}\big(s, x, T^{\dot{\mu}}_{s,0}(x,\Theta)\big) - \mathcal{V}^{\dot{\mu}}\big(s, x, T^{\dot{\nu}}_{s,0}(x,\Theta)\big)\, ds \\ &+ \int_0^t \mathcal{V}^{\dot{\mu}}\big(s, x, T^{\dot{\nu}}_{s,0}(x,\Theta)\big) - \mathcal{V}^{\dot{\nu}}\big(s, x, T^{\dot{\nu}}_{s,0}(x,\Theta)\big)\, ds.\end{aligned}$$

Recall that $\mathcal{V}^{\dot{\mu}}$ is Lipschitz in Θ with a constant smaller than $C(1+\|W\|_{L^\infty})$, as shown in the proof of Lemma 3.3.3. We thus have

$$\begin{aligned}\left|T^{\dot{\mu}}_{t,0}(x,\Theta) - T^{\dot{\nu}}_{t,0}(x,\Theta)\right| \leq C(1+\|W\|_{L^\infty}) &\int_0^t \left|T^{\dot{\mu}}_{s,0}(x,\Theta) - T^{\dot{\nu}}_{s,0}(x,\Theta)\right|\, ds \\ &+ \int_0^t \left|\mathcal{V}^{\dot{\mu}}\big(s, x, T^{\dot{\nu}}_{s,0}(x,\Theta)\big) - \mathcal{V}^{\dot{\nu}}\big(s, x, T^{\dot{\nu}}_{s,0}(x,\Theta)\big)\right|\, ds.\end{aligned}$$

Integrating in $\rho^x(d\Theta)$ and dx we find:

$$\begin{aligned}a(t) \leq C(1+\|W\|_{L^\infty}) &\int_0^t a(s)ds \\ &+ \int_0^t \int_\Omega \int_{\mathbb{T}\times\mathbb{R}} \left|\mathcal{V}^{\dot{\mu}}\big(s, x, T^{\dot{\nu}}_{s,0}(x,\Theta)\big) - \mathcal{V}^{\dot{\nu}}\big(s, x, T^{\dot{\nu}}_{s,0}(x,\Theta)\big)\right| \rho^x(d\Theta)dxds.\end{aligned}$$

For any arbitrary $\tilde{\Theta}$, we have:

$$\begin{aligned}\left|\mathcal{V}^{\dot{\mu}}(s,x,\tilde{\Theta}) - \mathcal{V}^{\dot{\nu}}(s,x,\tilde{\Theta})\right| &= \left|\int_\Omega W(x,y)\int_{\mathbb{T}\times\mathbb{R}} \sin(\xi-\tilde{\theta}-\beta)\left(\mu_s^y(d\xi,d\psi)-\nu_s^y(d\xi,d\psi)\right)dy\right| \\ &\leq C'\int_\Omega |W(x,y)|\, d(\mu_s^y,\nu_s^y)dy \\ &\leq C'\left\|W\right\|_{L^\infty}\bar{d}(\bar{\mu}_s,\bar{\nu}_s),\end{aligned}$$

where C' does not depend on $\left\|W\right\|_{L^\infty}$. We end up with:

$$a(t) \leq C(1+\left\|W\right\|_{L^\infty})\int_0^t a(s)ds + C'\left\|W\right\|_{L^\infty}\int_0^t \bar{d}(\bar{\mu}_s,\bar{\nu}_s)\ ds.$$

Applying Gronwall's Lemma 3.3.2, we obtain that:

$$a(t) \leq e^{C(1+\left\|W\right\|_{L^\infty})t}\int_0^t e^{-C(1+\left\|W\right\|_{L^\infty})s}C'\left\|W\right\|_{L^\infty}\bar{d}(\bar{\mu}_s,\bar{\nu}_s)\ ds.$$

Then we finally have:

$$\begin{aligned}d_\alpha(A\dot{\mu},A\dot{\nu}) &\leq C'\left\|W\right\|_{L^\infty}\sup_{t\in[0,T]} e^{-\alpha t+C(1+\left\|W\right\|_{L^\infty})t}\int_0^t e^{-C(1+\left\|W\right\|_{L^\infty})s}\bar{d}(\bar{\mu}_s,\bar{\nu}_s)\ ds \\ &\leq C'\left\|W\right\|_{L^\infty} d_\alpha(\dot{\mu},\dot{\nu})\sup_{t\in[0,T]} e^{-\alpha t+C(1+\left\|W\right\|_{L^\infty})t}\int_0^t e^{-C(1+\left\|W\right\|_{L^\infty})s+\alpha s}\ ds \\ &\leq d_\alpha(\dot{\mu},\dot{\nu})\frac{C'\left\|W\right\|_{L^\infty}}{\alpha - C(1+\left\|W\right\|_{L^\infty})},\end{aligned}$$

using that $\bar{d}(\bar{\mu}_s,\bar{\nu}_s) \leq e^{\alpha s}d_\alpha(\dot{\mu},\dot{\nu})$. Choosing $\alpha > 2C'\left\|W\right\|_{L^\infty} + C(1+\left\|W\right\|_{L^\infty})$ concludes, using Banach's fixed point theorem. □

Now that we have solutions to (FP) for any initial data $\bar{\rho}$, any graphon W and any frequencies ω, there are two things left to do in order to compare the solutions of (KMn) and the solution of the mean field equation (MF): study the dependence of the solutions of (FP) with respect to the parameters $\bar{\rho}$, W, ω, and show that the solutions of (MF) are also solutions of (FP). This is the purpose of the two next sections.

3.3.3 Continuous dependence of the solutions with respect to the data

In this section we prove that the solutions of (FP) depend continuously on the initial data, the graphon W and the frequencies ω. The proofs will be very similar to that of Lemma 3.3.6. We start by studying the dependence with respect to the initial data.

Lemma 3.3.7. *Let $\dot{\mu}$ and $\dot{\nu}$ be the unique solutions to (FP) with the same W and ω and with initial data $\bar{\mu}_0$, $\bar{\nu}_0$. For $\alpha > C_1(1+\left\|W\right\|_{L^\infty})$, we have:*

$$d^\alpha(\dot{\mu},\dot{\nu}) \leq C_2\bar{d}(\bar{\mu}_0,\bar{\nu}_0),$$

with a constant C_2 depending on T, D, d, $\left\|W\right\|_{L^\infty}$.

Proof. We write $A^{\bar{\mu}_0} = A[\bar{\mu}_0, \omega, W]$ and $A^{\bar{\nu}_0} = A[\bar{\nu}_0, \omega, W]$. Since $\dot{\mu}$ and $\dot{\mu}$ are fixed points of $A^{\bar{\mu}_0}$ and $A^{\bar{\nu}_0}$, we can write:

$$\begin{aligned} d_\alpha(\dot{\mu}, \dot{\nu}) &= d_\alpha(A^{\bar{\mu}_0}\dot{\mu}, A^{\bar{\nu}_0}\dot{\nu}) \\ &\leq d_\alpha(A^{\bar{\mu}_0}\dot{\mu}, A^{\bar{\mu}_0}\dot{\nu}) + d_\alpha(A^{\bar{\mu}_0}\dot{\nu}, A^{\bar{\nu}_0}\dot{\nu}) \\ &\leq \frac{1}{2} d_\alpha(\dot{\mu}, \dot{\nu}) + d_\alpha(A^{\bar{\mu}_0}\dot{\nu}, A^{\bar{\nu}_0}\dot{\nu}), \end{aligned}$$

where we used that for $\alpha > C_1(1 + \|W\|_{L^\infty})$, $A^{\bar{\mu}_0}$ is a $\frac{1}{2}$-contraction. Thus

$$d_\alpha(\dot{\mu}, \dot{\nu}) \leq 2 d_\alpha(A^{\bar{\mu}_0}\dot{\nu}, A^{\bar{\nu}_0}\dot{\nu}).$$

We simply have to control the right hand side:

$$\bar{d}(\overline{(A^{\bar{\mu}_0}\dot{\nu})}_t, \overline{(A^{\bar{\nu}_0}\dot{\nu})}_t) = \int_\Omega \sup_{f \in BL} \left| \int_{\mathbb{T}\times\mathbb{R}} f(T^{\dot{\nu}}_{t,0}(x,\Theta))\mu_0^x(d\Theta) - \int_{\mathbb{T}\times\mathbb{R}} f(T^{\dot{\nu}}_{t,0}(x,\Theta))\nu_0^x(d\Theta) \right| dx,$$

using once again the formula for pushforward of measures. But for almost every x, the function $\Theta \mapsto f(T^{\dot{\nu}}_{t,0}(x,\Theta))$ is Lipschitz because f is and the flow $T^{\dot{\nu}}_{t,0}(x,\cdot)$ is. The Lipschitz constant of $T^{\dot{\nu}}_{t,0}(x,\cdot)$ depends on T and $\|W\|_{L^\infty}$ as $L = \sqrt{e^{TC(1+\|W\|_{L^\infty})}}$ (see the proof of Lemma 3.3.3), so $\Theta \mapsto L^{-1} f(T^{\dot{\nu}}_{t,0}(x,\Theta))$ has Lipschitz constant 1. This yields:

$$\begin{aligned} e^{-\alpha t}\bar{d}(\overline{(A^{\bar{\mu}_0}\dot{\nu})}_t, \overline{(A^{\bar{\nu}_0}\dot{\nu})}_t) &\leq e^{-\alpha t} \int_\Omega L d(\mu_0^x, \nu_0^x) dx \\ &\leq L\bar{d}(\bar{\mu}_0, \bar{\nu}_0) \end{aligned}$$

Choosing $C_2 = 2L$ concludes. □

We also have continuous dependence of the solution on the graphon:

Lemma 3.3.8. *Let $\dot{\mu}$ and $\dot{\nu}$ be the unique solutions to (FP) with the same ω and with initial data $\bar{\mu}_0 = \bar{\nu}_0$, but for graphons W and V respectively. For $\alpha > C_1(1 + \|W\|_{L^\infty})$, we have:*

$$d_\alpha(\dot{\mu}, \dot{\nu}) \leq C_3 d_\square(W, V)$$

with a constant C_3 depending on T, D, d, $\|V\|_{L^\infty}$. Here the labelled cut distance $d_\square$ on graphons is given by:

$$d_\square(W, V) = \sup_{S,T \subset \Omega} \left| \int_S \int_T W(x,y) - V(x,y)\, dx\, dy \right|.$$

Proof. We use the following notations for the operators: $A^W = A[\bar{\mu}_0, \omega, W]$ and $A^V = A[\bar{\mu}_0, \omega, V]$. We will also use brackets, for instance $T^{\dot{\nu}}_{t,0}[W]$, $\mathcal{V}^{\dot{\nu}}[W]$, to precise which graphon the flow and velocity fields depend on. The computations are definitively not elegant, we advise a reader in a hurry to mainly track the dependencies in W and V. We start with the same trick as in the previous proof:

$$\begin{aligned} d_\alpha(\dot{\mu}, \dot{\nu}) &= d_\alpha(A^W\dot{\mu}, A^V\dot{\nu}) \\ &\leq d_\alpha(A^W\dot{\mu}, A^W\dot{\nu}) + d_\alpha(A^W\dot{\nu}, A^V\dot{\nu}) \\ &\leq \frac{1}{2} d_\alpha(\dot{\mu}, \dot{\nu}) + d_\alpha(A^W\dot{\nu}, A^V\dot{\nu}), \end{aligned}$$

which gives

$$d_\alpha(\dot\mu,\dot\nu) \le 2d_\alpha(A^W\dot\nu, A^V\dot\nu).$$

We now need to control:

$$\begin{aligned}\bar d(\overline{(A^W\dot\nu)}_t, \overline{(A^V\dot\nu)}_t) &= \int_\Omega \sup_{f\in BL}\left|\int_{\mathbb{T}\times\mathbb{R}} \big(f(T^{\dot\nu}_{t,0}[W](x,\Theta)) - f(T^{\dot\nu}_{t,0}[V](x,\Theta))\big)\mu_0^x(d\Theta)\right| dx\\ &\le \int_\Omega\int_{\mathbb{T}\times\mathbb{R}} \left|T^{\dot\nu}_{t,0}[W](x,\Theta) - T^{\dot\nu}_{t,0}[V](x,\Theta)\right| \mu_0^x(d\Theta)dx.\end{aligned}$$

Call this last integral $b(t)$. Using the definition of the flows,

$$b(t) \le \int_0^t\int_\Omega\int_{\mathbb{T}\times\mathbb{R}} \left|\mathcal{V}^{\dot\nu}[W](s,x,T^{\dot\nu}_{s,0}[W](x,\Theta)) - \mathcal{V}^{\dot\nu}[V](s,x,T^{\dot\nu}_{s,0}[V](x,\Theta))\right| \mu_0^x(d\Theta)dxds.$$

We cut the integrand in two parts, one with different fields but same flow and one with different flows but same field:

$$\begin{aligned}&\left|\mathcal{V}^{\dot\nu}[W](s,x,T^{\dot\nu}_{s,0}[W](x,\Theta)) - \mathcal{V}^{\dot\nu}[V](s,x,T^{\dot\nu}_{s,0}[V](x,\Theta))\right|\\ &\quad\le \left|\mathcal{V}^{\dot\nu}[W](s,x,T^{\dot\nu}_{s,0}[W](x,\Theta)) - \mathcal{V}^{\dot\nu}[V](s,x,T^{\dot\nu}_{s,0}[W](x,\Theta))\right|\\ &\quad+ \left|\mathcal{V}^{\dot\nu}[V](s,x,T^{\dot\nu}_{s,0}[W](x,\Theta)) - \mathcal{V}^{\dot\nu}[V](s,x,T^{\dot\nu}_{s,0}[V](x,\Theta))\right|.\end{aligned}$$

We use Lipschitz continuity of the velocity field to deal with the second term:

$$\begin{aligned}&\left|\mathcal{V}^{\dot\nu}[V](s,x,T^{\dot\nu}_{s,0}[W](x,\Theta)) - \mathcal{V}^{\dot\nu}[V](s,x,T^{\dot\nu}_{s,0}[V](x,\Theta))\right|\\ &\quad\le C(1+\|V\|_{L^\infty})\left|T^{\dot\nu}_{s,0}[W](x,\Theta) - T^{\dot\nu}_{s,0}[V](x,\Theta)\right|.\end{aligned}$$

Note that this term integrated over $\mathbb{T}\times\mathbb{R}$ and Ω with measure $\mu_0^x(d\Theta)dx$ is

$$C(1+\|V\|_{L^\infty})b(s).$$

The first term can be rewritten, letting $\tilde\Theta = T^{\dot\nu}_{s,0}[W](x,\Theta)$, as:

$$\left|\mathcal{V}^{\dot\nu}[W](s,x,\tilde\Theta)) - \mathcal{V}^{\dot\nu}[V](s,x,\tilde\Theta)\right| = \left|\int_\Omega \big(W(x,y)-V(x,y)\big)\int_{\mathbb{T}\times\mathbb{R}} \sin(\xi-\tilde\theta-\beta)\mu_s^y(d\xi,d\psi)dy\right|.$$

We expand the sinus, a method adapted from [3]

$$\sin(\xi-\tilde\theta-\beta) = \frac{1}{2i}e^{i\xi}e^{-i(\tilde\theta+\beta)} - \frac{1}{2i}e^{-i\xi}e^{i(\tilde\theta+\beta)}.$$

Let us treat the first term, the second being analogous:

$$\begin{aligned}&\left|\int_\Omega \big(W(x,y)-V(x,y)\big)\int_{\mathbb{T}\times\mathbb{R}} \frac{1}{2i}e^{i\xi}e^{-i(\tilde\theta+\beta)}\mu_s^y(d\xi,d\psi)dy\right|\\ &\quad= \frac12\left|\int_\Omega \big(W(x,y)-V(x,y)\big)h(y)dy\right|,\end{aligned}$$

where $h(y) = \int_{\mathbb{T}\times\mathbb{R}} e^{i\xi}\mu_s^y(d\xi, d\psi)$ so that $\|h\|_{L^\infty} \leq 1$. Integrating over $\mathbb{T}\times\mathbb{R}$ and Ω, we find:

$$\begin{aligned}\frac{1}{2}\int_\Omega \int_{\mathbb{T}\times\mathbb{R}} \left|\int_\Omega \big(W(x,y) - V(x,y)\big)h(y)dy\right| \mu_0^x(d\Theta)dx \\ = \frac{1}{2}\int_\Omega \left|\int_\Omega \big(W(x,y) - V(x,y)\big)h(y)dy\right| dx \\ \leq \frac{1}{2}\sup_{|h|\leq 1} \left\|\int_\Omega (W(\cdot,y) - V(\cdot,y))h(y)dy\right\|_{L^1}.\end{aligned}$$

Using the fact that the $L^\infty \to L^1$ operator norm is equivalent to the cut distance $d_\square$ (see Lemma 3.5.5 or [47]), we obtain:

$$\frac{1}{2}\int_\Omega \int_{\mathbb{T}\times\mathbb{R}} \left|\int_\Omega \big(W(x,y) - V(x,y)\big)h(y)dy\right| \mu_0^x(d\Theta)dx \leq C' d_\square(W,V).$$

Coming back to the control of $b(t)$, we have:

$$b(t) \leq C(1 + \|V\|_{L^\infty})\int_0^t b(s)ds + 2C'\int_0^t d_\square(W,V)ds.$$

We apply Gronwall's Lemma 3.3.2 to get that

$$b(t) \leq \frac{C_3}{2} d_\square(W,V),$$

which in turn yields

$$e^{-\alpha t}\bar{d}(\overline{(A^W\dot{\nu})}_t, \overline{(A^V\dot{\nu})}_t) \leq e^{-\alpha t}\frac{C_3}{2} d_\square(W,V)$$

and the desired the result

$$d_\alpha(\dot{\mu}, \dot{\nu}) \leq C_3 d_\square(W,V).$$

□

We conclude this section with the analogous result for the frequencies ω:

Lemma 3.3.9. *Let $\dot{\mu}$ and $\dot{\nu}$ be the unique solutions to (FP) with the same W and with initial data $\bar{\mu}_0 = \bar{\nu}_0$, but for frequencies ω and ω' respectively. For $\alpha > C_1(1 + \|W\|_{L^\infty})$, we have:*

$$d_\alpha(\dot{\mu}, \dot{\nu}) \leq C_4 \left\|\omega - \omega'\right\|_{L^1}$$

with a constant C_4 depending on T, D, d, $\|W\|_{L^\infty}$.

Proof. We carry out the exact same proof as before. With straightforwardly adapted notations, the only difference is the control of the term with different fields but same flows:

$$\left|\mathcal{V}^{\dot{\nu}}[\omega](s,x,T^{\dot{\nu}}_{s,0}[\omega](x,\Theta)) - \mathcal{V}^{\dot{\nu}}[\omega'](s,x,T^{\dot{\nu}}_{s,0}[\omega](x,\Theta))\right|.$$

This time the expression is much simpler:

$$\left|\mathcal{V}^{\dot{\nu}}[\omega](s,x,T^{\dot{\nu}}_{s,0}[\omega](x,\Theta)) - \mathcal{V}^{\dot{\nu}}[\omega'](s,x,T^{\dot{\nu}}_{s,0}[\omega](x,\Theta))\right| = \left|\omega(x) - \omega'(x)\right|.$$

Integrating in Θ is trivial, and the integral in x yields the L^1 norm that appears in the result instead of the previous cut norm. □

We now entirely control the differences between solutions of the fixed point (FP) for different initial data, graphons and frequencies. We are all set to treat the convergence towards the mean field limit.

3.4 The mean field limit

3.4.1 From the fixed point to the mean field limit

We first show the equivalence between solving the fixed point (FP) and the mean field equation (MF) in a weak sense. The weak formulation of (MF) naturally stems by differentiating in time the fixed point relation. Indeed, consider $\dot{\mu}$ a solution of (FP), and let us fix a test function f and perform formal computations. We have

$$\int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\mu_t^x(d\Theta) = \int_{\mathbb{T}\times\mathbb{R}} f(T_{t,0}^{\dot{\mu}}(x,\Theta))\mu_0^x(d\Theta).$$

Differentiating with respect to time, we get:

$$\begin{aligned}\frac{d}{dt}\int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\mu_t^x(d\Theta) &= \int_{\mathbb{T}\times\mathbb{R}} \partial_t T_{t,0}^{\dot{\mu}}(x,\Theta)\cdot\nabla f(T_{t,0}^{\dot{\mu}}(x,\Theta))\mu_0^x(d\Theta)\\ &= \int_{\mathbb{T}\times\mathbb{R}} \mathcal{V}^{\dot{\mu}}(t,x,T_{t,0}^{\dot{\mu}}(x,\Theta))\cdot\nabla f(T_{t,0}^{\dot{\mu}}(x,\Theta))\mu_0^x(d\Theta)\\ &= \int_{\mathbb{T}\times\mathbb{R}} \mathcal{V}^{\dot{\mu}}(t,x,\Theta)\cdot\nabla f(\Theta)\mu_t^x(d\Theta).\end{aligned}$$

A very formal integration by parts would yield:

$$\int_{\mathbb{T}\times\mathbb{R}} \mathcal{V}^{\dot{\mu}}(t,x,\Theta)\cdot\nabla f(\Theta)\mu_t^x(d\Theta) = -\int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\ \mathrm{div}_\Theta\big(\mathcal{V}^{\dot{\mu}}(t,x,\Theta)\mu_t^x(d\Theta)\big),$$

and thus the equation

$$\partial_t\mu_t^x(d\Theta) + \mathrm{div}_\Theta\big(\mathcal{V}^{\dot{\mu}}(t,x,\Theta)\mu_t^x(d\Theta)\big) = 0,$$

which is exactly (MF) using the definition of the field $\mathcal{V}^{\dot{\mu}}$. We thus define weak solutions of (MF) as follows:

Definition 3.4.1. *We say that $\dot{\mu}\in\dot{\mathcal{M}}_T$ is a weak solution of (MF) if, for almost every $x\in\Omega$, we have that for any differentiable function on $\mathbb{T}\times\mathbb{R}$ with compact support and continuous bounded derivatives $f\in\mathcal{C}_c^1(\mathbb{T}\times\mathbb{R})$, the following property holds:*

The function $t\longmapsto\int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\mu_t^x(d\Theta)$ is continuously differentiable on $[0,T]$, and for all $t\in[0,T]$:

$$\frac{d}{dt}\int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\mu_t^x(d\Theta) = \int_{\mathbb{T}\times\mathbb{R}} \mathcal{V}^{\dot{\mu}}(t,x,\Theta)\cdot\nabla f(\Theta)\ \mu_t^x(d\Theta).$$

In view of the previous computations, we have equivalence between the weak formulation of (MF) and (FP), which gives well-posedness for the mean field limit. This is made rigorous in the following lemma:

Lemma 3.4.2. *For any initial condition $\bar{\mu}_0\in\bar{\mathcal{M}}$, there exists a unique weak solution to (MF) $\dot{\mu}$.*

Moreover, a distribution $\dot{\mu}\in\dot{\mathcal{M}}_T$ is a weak solution of (MF) if and only if it solves the fixed point equation (FP).

Finally, if $\bar{\mu}_0\in\bar{\mathcal{M}}$ is absolutely continuous and given by a density $\rho_0^x(\Theta)$, then the solution at any time t is given by a density $\rho_t^x(\Theta)$ that solves (MF) in the sense of distributions.

Proof. We first show that a solution $\dot{\mu}$ of (FP) with initial condition $\bar{\mu}_0$ is a weak solution of (MF). For almost every x, for any test function f, we have:

$$\int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\mu_t^x(d\Theta) = \int_{\mathbb{T}\times\mathbb{R}} f(T_{t,0}^{\dot{\mu}}(x,\Theta))\mu_0^x(d\Theta).$$

Let us justify that this is continuously differentiable in time. Thanks to the properties of the flow, the function $t \mapsto f(T_{t,0}^{\dot{\mu}}(x,\Theta))$ is continuously differentiable in time with derivative given by:

$$g(t,\Theta) = \mathcal{V}^{\dot{\mu}}(t,x,T_{t,0}^{\dot{\mu}}(x,\Theta)) \cdot \nabla f(T_{t,0}^{\dot{\mu}}(x,\Theta)).$$

Denote by $(\tilde{\theta}(t,\Theta),\tilde{\phi}(t,\Theta)) = T_{t,0}^{\dot{\mu}}(x,\Theta)$. Since f has compact support, $\tilde{\phi}(t,\Theta)$ is bounded for (t,Θ) in the support of g. Thus we can uniformly bound $\mathcal{V}^{\dot{\mu}}$ and, since ∇f is bounded as well, $g(t,\Theta) \leq C$ for some C for all (t,Θ). We can perform differentiation under the integral sign to get:

$$\begin{aligned}\frac{d}{dt}\int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\mu_t^x(d\Theta) &= \int_{\mathbb{T}\times\mathbb{R}} \mathcal{V}^{\dot{\mu}}(t,x,T_{t,0}^{\dot{\mu}}(x,\Theta)) \cdot \nabla f(T_{t,0}^{\dot{\mu}}(x,\Theta))\mu_0^x(d\Theta) \\ &= \int_{\mathbb{T}\times\mathbb{R}} \mathcal{V}^{\dot{\mu}}(t,x,\Theta) \cdot \nabla f(\Theta)\mu_t^x(d\Theta).\end{aligned}$$

Consider now $\dot{\mu}$ a weak solution of (MF). The defining property of weak solution holds almost everywhere in x. For such x, consider a test function f, and let $g(t,\Theta) = f(T_{0,t}^{\dot{\mu}}(x,\Theta))$. Since for $s < t$, we have $T_{0,t}^{\dot{\mu}} = T_{0,s}^{\dot{\mu}} \circ T_{s,t}^{\dot{\mu}}$, we get the identity

$$g(t,\Theta) = g(s, T_{s,t}^{\dot{\mu}}(x,\Theta))$$

that we can differentiate in time to obtain:

$$\begin{aligned}\frac{d}{dt}g(t,\Theta) &= \frac{d}{dt}g(s,T_{s,t}^{\dot{\mu}}(x,\Theta))|_{s=t} \\ &= \nabla_\Theta g(t,\Theta) \cdot \frac{d}{dt}T_{s,t}^{\dot{\mu}}(x,\Theta)|_{s=t} \\ &= \nabla_\Theta g(t,\Theta) \cdot (-\mathcal{V}^{\dot{\mu}}(t,x,\Theta)).\end{aligned}$$

We used that $\frac{d}{dt}T_{s,t}^{\dot{\mu}}(x,\Theta)|_{s=t} = -\mathcal{V}^{\dot{\mu}}(t,x,\Theta)$, since the inverse flow is the flow obtained by considering the opposite velocity field $-\mathcal{V}^{\dot{\mu}}$. Plugging this we get:

$$\begin{aligned}\frac{d}{dt}\int_{\mathbb{T}\times\mathbb{R}} g(t,\Theta)\mu_t^x(d\Theta) &= \frac{\partial}{\partial t}\int_{\mathbb{T}\times\mathbb{R}} g(t,\Theta)\mu_s^x(d\Theta)|_{s=t} + \frac{\partial}{\partial s}\int_{\mathbb{T}\times\mathbb{R}} g(t,\Theta)\mu_s^x(d\Theta)|_{s=t} \\ &= \int_{\mathbb{T}\times\mathbb{R}} \nabla_\Theta g(t,\Theta) \cdot (-\mathcal{V}^{\dot{\mu}}(t,x,\Theta))\mu_t^x + \int_{\mathbb{T}\times\mathbb{R}} \nabla_\Theta g(t,\Theta) \cdot \mathcal{V}^{\dot{\mu}}(t,x,\Theta)\mu_t^x \\ &= 0.\end{aligned}$$

The first differentiation under the integral is justified the same way as before, the second is by definition of weak solutions. We thus obtain for any test function f:

$$\int_{\mathbb{T}\times\mathbb{R}} f(T_{0,t}^{\dot{\mu}}(x,\Theta))\mu_t^x(d\Theta) = \int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\mu_0^x(d\Theta),$$

which is a reformulation of the fixed point equation.

Existence and uniqueness for (MF) results from the well-posedness of (FP) (Lemma 3.3.6). It remains to prove the part about absolutely continuous measures. Suppose the initial condition given by a density $\rho_0^x(\Theta)$. Since the flow $T_{t,0}^{\dot\mu}$ is Lipschitz in Θ, it admits almost everywhere a derivative and we can make use of the change of variables formula for Lipschitz maps:

$$\begin{aligned}\int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\mu_t^x(d\Theta) &= \int_{\mathbb{T}\times\mathbb{R}} f(T_{t,0}^{\dot\mu}(x,\Theta))\mu_0^x(d\Theta)\\ &= \int_{\mathbb{T}\times\mathbb{R}} f(T_{t,0}^{\dot\mu}(x,\Theta))\rho_0^x(\Theta)d\Theta\\ &= \int_{\mathbb{T}\times\mathbb{R}} f(\Theta)\rho_0^x(T_{0,t}^{\dot\mu}(x,\Theta))\left|\det D_\Theta T_{0,t}^{\dot\mu}(x,\Theta)\right| d\Theta.\end{aligned}$$

Thus μ_t^x is absolutely continuous with density $\rho_t^x(\Theta) = \rho_0^x(T_{0,t}^{\dot\mu}(x,\Theta))\left|\det D_\Theta T_{0,t}^{\dot\mu}(x,\Theta)\right|$. That ρ verifies (MF) in the sense of distribution is a consequence of $\dot\mu$ solving the weak formulation. □

3.4.2 Convergence to the mean field limit

We have all the necessary results to prove convergence of the solutions of the Kuramoto model as $n\to\infty$ to a solution of the mean field equation. Both solutions of (KMn) and (MF) solve the fixed point equation, for which we proved continuous dependence on the initial data, the intrinsic frequencies ω and the graphon W.

From our previous results we deduce the following key estimate:

Theorem 3.4.3. *For every $n\in\mathbb{N}^*$, consider the empirical measures $\dot\mu^n$ of the Kuramoto model (KMn) with weights W_{ij}^n given by the graphon $W^n\in L^\infty(\Omega^2)$:*

$$W^n(x,y) = \frac{W_{ij}^n}{n|P_j^n|} \text{ for } x\in P_i^n,\ y\in P_j^n$$

and with intrinsic frequencies given by $\omega^n\in L^1(\Omega)$:

$$\omega^n(x) = \omega_i^n \text{ for } x\in P_i^n.$$

Let $\dot\mu\in\dot{\mathcal{M}}_T$ be the solution to the mean field equation (MF) with initial data $\mu_0^x\in\bar{\mathcal{M}}$, graphon $W\in L^\infty(\Omega^2)$, and frequencies $\omega\in L^1(\Omega)$.

Suppose that the L^∞ norm of the graphons remains bounded :

$$\forall n,\ \|W^n\|_{L^\infty} < M.$$

Then, there exists $C,c>0$ depending on D, d, M, T, $\|W\|_{L^\infty}$ such that, for $\alpha>c$, the following inequality holds:

$$d_\alpha(\dot\mu^n,\dot\mu) \le C\big(\bar d(\overline{(\mu^n)}_0,\overline{\mu}_0) + d_\square(W^n,W) + \|\omega^n-\omega\|_{L^1}\big).$$

Proof. The mean field solution $\dot{\mu}$ satisfies (FP) with parameters W, ω and initial condition $\overline{\mu}_0$ thanks to Lemma 3.4.2. Up to changing M, we can suppose $\|W\|_{L^\infty} < M$. By Lemma 3.3.4 the empirical measures $\dot{\mu}^n$ satisfy (FP) with frequencies ω^n and graphon W^n. We apply Lemmas 3.3.7, 3.3.8, 3.3.9 in a row to obtain that for $\alpha > c = C_1(1+M)$,

$$d_\alpha(\dot{\mu}^n, \dot{\mu}) \leq C_2 \bar{d}(\overline{(\mu^n)}_0, \overline{\mu}_0) + C_3 d_\square(W^n, W) + C_4 \|\omega^n - \omega\|_{L^1},$$

by comparing the respective solutions of (FP) when changing only one parameter at a time. Letting $C = \max(C_2, C_3, C_4)$ concludes. □

The question that now arises is to know how one verifies in practice the convergence of the different terms in the estimate given by Theorem 3.4.3. We first give a way to easily construct converging graph sequences. One way to approximate a given limit graphon W by a finite graph on n vertices is by setting

$$W^n(x,y) = \frac{1}{|P_i^n \times P_j^n|} \int_{P_i^n \times P_j^n} W(x',y')dx'dy' \text{ for } x \in P_i^n, y \in P_j^n.$$

Under reasonable regularity assumptions on the partition $(P_i^n)_{1\leq i\leq n}$, this definition correctly approximates the graphon W. We can also build frequencies ω^n with the same approach.

Lemma 3.4.4. *Suppose the partition $(P_i^n)_{1\leq i\leq n}$ satisfies*

$$\delta^n = \sup\{|x - x'| \mid x, x' \in P_i^n, 1 \leq i \leq n\} \xrightarrow[n\to\infty]{} 0.$$

Let $\omega \in L^1(\Omega)$, and define, for $n \in \mathbb{N}^*$, ω^n as

$$\omega^n(x) = \frac{1}{|P_i^n|} \int_{P_i^n} \omega(x')dx' \text{ for } x \in P_i^n.$$

Then $\|\omega^n\|_{L^1} \leq \|\omega\|_{L^1}$ and

$$\|\omega^n - \omega\|_{L^1} \xrightarrow[n\to\infty]{} 0.$$

Likewise, let $W \in L^1(\Omega^2)$, and define W^n as

$$W^n(x,y) = \frac{1}{|P_i^n \times P_j^n|} \int_{P_i^n \times P_j^n} W(x',y')dx'dy' \quad \text{for } x \in P_i^n,\ y \in P_j^n.$$

Then $\|W^n\|_{L^1} \leq \|W\|_{L^1}$ and

$$\|W^n - W\|_{L^1} \xrightarrow[n\to\infty]{} 0.$$

Proof. We prove the result for ω, it is exactly the same for W. First

$$\begin{aligned}
\|\omega^n\|_{L^1} &= \sum_{i=1}^n \left| \int_{P_i^n} \omega(x)dx \right| \\
&\leq \sum_{i=1}^n \int_{P_i^n} |\omega(x)|\, dx \\
&= \|\omega\|_{L^1}
\end{aligned}$$

Now pick $\varepsilon > 0$. By density, there exists a Lipschitz function g such that $\|g - \omega\|_{L^1} < \varepsilon/3$. We then also have for all n:

$$\|g^n - \omega^n\|_{L^1} \leq \|g - \omega\|_{L^1} < \frac{\varepsilon}{3}$$

If K is a Lipschitz constant of g, we have:

$$\begin{aligned}
\|g^n - g\|_{L^1} &= \sum_{i=1}^{n} \int_{P_i^n} \left| \frac{1}{|P_i^n|} \int_{P_i^n} g(y)dy - g(x) \right| dx \\
&= \sum_{i=1}^{n} \int_{P_i^n} \left| \frac{1}{|P_i^n|} \int_{P_i^n} (g(y) - g(x))dy \right| dx \\
&\leq \sum_{i=1}^{n} \int_{P_i^n} K\delta^n \\
&= K\delta^n
\end{aligned}$$

For n sufficiently large, we have $K\delta^n < \varepsilon/3$. Then for such n,

$$\|\omega^n - \omega\|_{L^1} \leq \|\omega^n - g^n\|_{L^1} + \|g^n - g\|_{L^1} + \|g - \omega\|_{L^1} < \varepsilon.$$

□

We now turn to the convergence of initial data, in the deterministic case. The case of random initial data will be more subtle and will be treated in the next section.

Lemma 3.4.5. *Suppose the partition $(P_i^n)_{1 \leq i \leq n}$ satisfies*

$$\delta^n = \sup\{\left|x - x'\right| \mid x, x' \in P_i^n, 1 \leq i \leq n\} \xrightarrow[n \to \infty]{} 0.$$

Let $\Theta_0 \in L^1(\Omega, \mathbb{T} \times \mathbb{R})$ and define initial conditions for (KMn) by

$$\Theta_i^n(0) = (\theta_i^n(0), \frac{d}{dt}\theta_i^n(0)) = \frac{1}{|P_i^n|} \int_{P_i^n} \Theta_0(x)dx$$

Then the empirical initial measures $\overline{(\mu^n)}_0$ converge in the $\bar{d}$ metric:

$$\bar{d}(\overline{(\mu^n)}_0, \overline{\mu}_0) \xrightarrow[n \to \infty]{} 0,$$

and $\bar{\mu}_0$ is given by

$$\mu_0^x = \delta_{\Theta_0(x)}$$

Proof. We write:

$$\begin{aligned}
\bar{d}(\overline{(\mu^n)}_0, \overline{\mu}_0) &= \int_\Omega d((\mu^n)_0^x,\ \mu_0^x)dx \\
&= \sum_{i=1}^n \int_{P_i^n} d((\mu^n)_0^x,\ \mu_0^x)dx \\
&= \sum_{i=1}^n \int_{P_i^n} \sup_{f\in BL} |f(\Theta_i^n(0)) - f(\Theta_0(x))|\, dx \\
&\leq \sum_{i=1}^n \int_{P_i^n} |\Theta_i^n(0) - \Theta_0(x)|\, dx \\
&= \|\Theta^n - \Theta\|_{L^1}
\end{aligned}$$

by using the notations of Lemma 3.4.4. Adapting this same lemma, we claim that

$$\|\Theta^n - \Theta\|_{L^1} \xrightarrow[n\to\infty]{} 0.$$

□

Building initial conditions, graphs, and frequencies in this way ensures convergence of the Kuramoto model to the corresponding mean field limit:

Theorem 3.4.6. *Let $\omega \in L^1(\Omega)$, $W \in L^\infty(\Omega)$, and $\Theta_0 \in L^1(\Omega, \mathbb{T}\times\mathbb{R})$. Let ω^n, W^n and initial conditions $(\theta_i^n(0), \frac{d}{dt}\theta_i^n(0))$ be defined as in Lemmas 3.4.4 and 3.4.5. Let $\dot{\mu}^n$ be the empirical measure of the Kuramoto model (KMn) with frequencies and graph defined as in Theorem 3.4.3. Let $\dot{\mu}$ be the solution to (MF) with initial data $\mu_0^x = \delta_{\Theta_0(x)}$, frequencies ω and graphon W.*

Suppose the partition $(P_i^n)_{1\leq i\leq n}$ satisfies

$$\delta^n = \sup\{\left|x - x'\right| \ |\ x, x' \in P_i^n, 1 \leq i \leq n\} \xrightarrow[n\to\infty]{} 0.$$

Then $\dot{\mu}^n$ converges to $\dot{\mu}$: there exists $c > 0$ such that for $\alpha > c$,

$$d_\alpha(\dot{\mu}^n, \dot{\mu}) \xrightarrow[n\to\infty]{} 0.$$

Moreover, $\dot{\mu}$ is given by

$$\mu_t^x = \delta_{\Theta(t,x)}$$

with $\Theta(t,x) = T_{t,0}^{\dot{\mu}}(x, \Theta_0(x))$.

Proof. We apply theorem 3.4.3, by remarking that the L^∞ norm of W^n remains bounded since $\|W^n\|_{L^\infty} \leq \|W\|_{L^\infty}$. We deduce that for $\alpha > c > 0$,

$$d_\alpha(\dot{\mu}^n, \dot{\mu}) \leq C\big(\bar{d}(\overline{(\mu^n)}_0, \overline{\mu}_0) + d_\square(W^n, W) + \|\omega^n - \omega\|_{L^1}\big).$$

All the terms in this estimate converge to 0 by applying Lemmas 3.4.4 and 3.4.5, since the cut distance is controlled by the L^1 norm. Finally, since $\dot{\mu}$ satisfies the fixed point equation, we have:

$$\begin{aligned}
\int_\Omega f(\Theta)\mu_t^x(d\Theta) &= \int_\Omega f(T_{t,0}^{\dot{\mu}}(x,\Theta))\mu_0^x(d\Theta) \\
&= f(T_{t,0}^{\dot{\mu}}(x, \Theta_0(x))).
\end{aligned}$$

□

The hypothesis of deterministic initial conditions is not the most natural one to make. In practice, we do not have much information on the initial conditions of power systems, so a more natural choice is to draw them randomly. The next section shows almost sure convergence of the empirical measure in that case.

3.4.3 Convergence for random initial data

The case of random initial data will require some work because we cannot directly apply results such as the Law of Large Numbers. Indeed, what we need is the convergence of the initial data in the $\bar{d}$ distance, because then Theorem 3.4.3 yields convergence on finite time intervals (provided convergence of the graphons and frequencies). This is not straightforward to obtain, and in fact we will need to weaken this metric a bit. To see why, consider the case where the empirical measure is an averaged sum of Dirac masses (as in the standard all-to-all coupled case for instance, so we can forget the dependence in x):

$$\mu_t^n = \frac{1}{n}\sum_{i=1}^{n} \delta_{\theta_i(t)}.$$

If the initial data ($\theta_i(0)$ is sampled independantly from a law ρ_0, the Law of Large Numbers gives direct results: integrating the initial empirical measure μ_0^n against a test function f gives an averaged sum of random variables:

$$\int f d\mu_0^n = \frac{1}{n}\sum_{i=1}^{n} f(\theta_i(0)).$$

which then converges almost surely to the expectancy, which is the integral of f against the law ρ_0. We thus have weak convergence of the initial empirical measure to ρ_0. However, in our case, the empirical measure at a point x is only a single Dirac mass rather than an empirical mean, so it has no chance to converge to the law ρ_0 in the bounded Lipschitz distance. Since the $\bar{d}$ metric is the integral over Ω of the bounded Lipschitz distance at x, we cannot directly obtain the desired convergence in $\bar{d}$ metric. How we solve this is by artificially introducing a locally averaged version of the empirical measure, to which we apply the Law of Large Numbers ; while making sure that it remains close enough to the empirical measure. This will require to weaken the $\bar{d}$ metric a bit.

The techniques are once again adapted from [28]. We will denote by $d_{\Omega\times\mathbb{T}\times\mathbb{R}}$ the Bounded Lipschitz distance on the set of probability measures $\mathcal{M}(\Omega\times\mathbb{T}\times\mathbb{R})$, that is

$$d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{\mu},\overline{\nu}) = \sup_{f\in BL(\Omega\times\mathbb{T}\times\mathbb{R})} \left| \int_{\Omega\times\mathbb{T}\times\mathbb{R}} f(d\overline{\mu} - d\overline{\nu}) \right|.$$

Note that we can see elements of $\bar{\mathcal{M}}$ as probabilities on $\Omega\times\mathbb{T}\times\mathbb{R}$ by setting for any Borel set A:

$$\overline{\mu}(A) = \int_\Omega \int_{\mathbb{T}\times\mathbb{R}} \mathbf{1}_A(x,\Theta)\mu^x(d\Theta)dx.$$

We will consider N oscillators and suppose we can write N as a product of two integer parameters $N = nm$. Instead of using an arbitrary partition P_i^N, we will refine the partition

P_i^n: consider subsets $P_{i,j}^{n,m}$ for $1 \leq i \leq n$ and $1 \leq j \leq m$ such that

$$P_i^n = \bigsqcup_{j=1}^{m} P_{i,j}^{n,m}.$$

We also consider points $x_{i,j}^{n,m} \in P_{i,j}^{n,m}$. On the partitions we make the following hypothesis:
1) We suppose the diameters of the partition P_i^n uniformly go to zero:

$$\delta^n = \sup\{\left|x - x'\right| \mid x, x' \in P_i^n, 1 \leq i \leq n\} \xrightarrow[n\to\infty]{} 0.$$

2) We suppose the diameter of the finer partition also uniformly go to zero at n fixed:

$$\delta^{n,m} = \sup\{\left|x - x'\right| \mid x, x' \in P_{i,j}^{n,m}, 1 \leq i \leq n, 1 \leq m \leq j\} \xrightarrow[m\to\infty]{} 0.$$

3) Finally, we suppose for simplicity that $\left|P_{i,j}^{n,m}\right| = \frac{1}{N}$ for all i, j.

Remark 3.4.7. *This last assumption could be relaxed to*

$$\sup_i \left| \frac{1}{n\left|P_i^n\right|} - 1 \right| \xrightarrow[n\to\infty]{} 0,$$

$$\sup_{i,j} \left| \frac{1}{nm\left|P_{i,j}^{n,m}\right|} - 1 \right| \xrightarrow[m\to\infty]{} 0,$$

it simply adds error terms in the following computations that we avoid for the sake of clarity. It also gets rid of the correction factor $\frac{1}{nP_i^n}$ that we encountered before in the definition of graphons.

We fix a function $\rho_0 = \rho_0(x, \Theta)$ such that ρ_0 is uniformly continuous in x and for all x, $\rho_0(x, \cdot)$ is a probability density on $\mathbb{T} \times \mathbb{R}$. We consider an array of initial conditions given by the random variables $(A_{i,j}^{n,m})_{1\leq i\leq n, 1\leq j\leq m}$, drawn independently, with law $\rho_0(x_{i,j}^{n,m}, \cdot)$

Since we focus on the convergence of initial data, we will make simple assumptions for the convergence of graphons and frequencies, and consider the case we studied before. Let $\omega \in L^1(\Omega)$ and $W \in L^\infty(\Omega)$, we define:

$$W^{n,m}(x, y) = \frac{1}{|P_{i,j}^{n,m} \times P_{k,l}^{n,m}|} \int_{P_{i,j}^{n,m} \times P_{k,l}^{n,m}} W(x', y')dx'dy' \quad \text{for } x \in P_{i,j}^{n,m},\ y \in P_{k,l}^{n,m}.$$

and its straightforward adaptation $\omega^{n,m}$.

The Kuramoto model on N oscillators $(\theta_{i,j}^{n,m})_{i,j}$ in the new notation writes:

$$\begin{aligned}(\text{KMnm}): \ (\frac{d^2}{dt^2} + D\frac{d}{dt})\theta_{i,j}^{n,m}(t) &= \omega^{n,m}(x_{i,j}^{n,m}) \\ &+ \frac{1}{N}\sum_{k=1}^{n}\sum_{l=1}^{m} W^{n,m}(x_{i,j}^{n,m}, x_{k,l}^{n,m}) \sin(\theta_{k,l}^{n,m}(t) - \theta_{i,j}^{n,m}(t) - \beta)\end{aligned}$$

The result of this section is the following:

Theorem 3.4.8. *Consider $\dot{\mu}^{n,m}$ the empirical measure of (KMmn) with random initial conditions $(A^{n,m}_{i,j})_{i,j}$ stated above, graphon $W^{n,m}$ and frequencies $\omega^{n,m}$.*

Let $\dot{\mu}$ be the solution of (MF) with initial data $\overline{\mu}_0$ given by

$$\mu_0^x = \rho_0(x,\Theta)d\Theta$$

and with graphon W and frequencies ω. Then, almost surely, for any $\varepsilon > 0$, there exists n_0 such that, for any $n \geq n_0$, there exists m_0 so that for $m \geq m_0$:

$$\sup_{t\in[0,T]} d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{(\mu^{n,m})}_t, \overline{\mu}_t) < \varepsilon.$$

The proof relies on the following averaged empirical measure:

$$(\nu^{n,m})^x_t = \frac{1}{m}\sum_{j=1}^{m} \delta_{(\tilde{\theta}^{n,m}_{i,j}(t), \frac{d}{dt}\tilde{\theta}^{n,m}_{i,j}(t))} \text{ for } x \in P^n_i$$

where $\tilde{\theta}^{n,m}_{i,j}$ is the solution of (KMnm) with same initial data but with graphons and frequencies replaced by the version averaged on the rougher partition P^n_i:

$$\tilde{W}^n(x,y) = \frac{1}{|P^n_i \times P^n_k|}\int_{P^n_i\times P^n_k} W(x',y')dx'dy' \quad \text{for } x \in P^n_i,\ y \in P^n_k.$$

and the analogously defined frequencies $\tilde{\omega}^n$. We will refer to those functions as averaged ones (even if $W^{n,m}$ and $\omega^{n,m}$ are also technically averaged versions of W and ω).

The proof will be divided in three steps: First we show that the empirical measure is close to the averaged version in the $d_{\Omega\times\mathbb{T}\times\mathbb{R}}$ metric for large n. We then show convergence as $m \to \infty$ of the averaged measure to a measure $\dot{\rho^n}$, using the law of large numbers, in the $\bar{d}$ metric. We finally show the convergence of $\dot{\rho^n}$ to $\dot{\mu}$.

We begin by showing we can replace the graphon and frequency by their averaged versions.

Lemma 3.4.9. *Let $\dot{\gamma}^{n,m}$ be the usual empirical measure of the Kuramoto model (KMnm) with graphons and frequencies replaced by $\tilde{W}^n$, $\tilde{\omega}^n$, and same initial data $(A^{n,m}_{i,j})_{i,j}$. For any $\epsilon > 0$, there exists n_1 such that, for all $n \geq n_1$, for all m, and for $\alpha > c > 0$:*

$$d_\alpha(\dot{\gamma}^{n,m}, \dot{\mu}^{n,m}) \leq \epsilon$$

As before, the constant c depends on d, D, $\|W\|_{L^\infty}$ but not on ϵ.

Proof. The empirical measures $\dot{\gamma}^{n,m}$ and $\dot{\mu}^{n,m}$ both satisfy the fixed point equation (FP). We invoke Lemmas 3.3.8 and 3.3.9 to get:

$$d_\alpha(\dot{\gamma}^{n,m}, \dot{\mu}^{n,m}) \leq C(\|\omega^{n,m} - \tilde{\omega}^n\|_{L^1} + d_\square(W^{n,m}, \tilde{W}^n))$$

Using Lemma 3.4.4 with the partition $P^{n,m}_{i,j}$, we get that $\|\omega^{n,m} - \omega\|_{L^1} \xrightarrow[n\to\infty]{} 0$. This convergence is uniform in m: looking at the proof of Lemma 3.4.4, we see that we need to control the diameter $\delta^{n,m}$ of the partition $P^{n,m}_{i,j}$, which is smaller than the diameter δ^n for all m.

Moreover, since $\tilde{\omega}^n$ is the piecewise constant approximation of ω, this time over the partition P^n_i, we have once again thanks to Lemma 3.4.4,

$$\|\omega - \tilde{\omega}^n\|_{L^1} \xrightarrow[n\to\infty]{} 0.$$

The same argument for the graphon concludes. □

We now relate $\dot{\gamma}^{n,m}$ and the averaged empirical measure $\dot{\nu}^{n,m}$.

Lemma 3.4.10. *For any $\epsilon > 0$, there exists n_2 such that, for all $n \geq n_2$, for all m,*

$$\sup_{t\in[0,T]} d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{\gamma}_t^{n,m}, \overline{\nu}_t^{n,m}) \leq \epsilon$$

Proof. The proof is a straightforward estimate. For any t, we have:

$$\begin{aligned}
&d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{\gamma}_t^{n,m}, \overline{\nu}_t^{n,m}) \\
&= \sup_{f\in BL(\Omega\times\mathbb{T}\times\mathbb{R})} \left| \sum_{i=1}^{n} \int_{P_i^n} \frac{1}{m} \sum_{j=1}^{m} f(x, \tilde{\Theta}_{i,j}^{n,m}(t))dx - \sum_{i=1}^{n}\sum_{j=1}^{m} \int_{P_{i,j}^{n,m}} f(y, \tilde{\Theta}_{i,j}^{n,m}(t))dy \right| \\
&= \sup_{f\in BL(\Omega\times\mathbb{T}\times\mathbb{R})} \frac{1}{m} \left| \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{l=1}^{m} \int_{P_{i,l}^{n,m}} f(x, \tilde{\Theta}_{i,j}^{n,m}(t))dx - \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{l=1}^{m} \int_{P_{i,j}^{n,m}} f(y, \tilde{\Theta}_{i,j}^{n,m}(t))dy \right|
\end{aligned}$$

We partitioned P_i^n in the first term and artificially added a sum over l in the second term. Then, using that $|P_{i,l}^{n,m}| = \frac{1}{nm}$ and then that f is Lipschitz,

$$\begin{aligned}
&d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{\mu}_t^{n,m}, \overline{\nu}_t^{n,m}) \\
&= \sup_{f\in BL(\Omega\times\mathbb{T}\times\mathbb{R})} \frac{1}{m} \left| \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{l=1}^{m} nm \int_{P_{i,l}^{n,m}} \int_{P_{i,j}^{n,m}} \big(f(x, \tilde{\Theta}_{i,j}^{n,m}(t)) - f(y, \tilde{\Theta}_{i,j}^{n,m}(t))\big)dydx \right| \\
&\leq n \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{l=1}^{m} \int_{P_{i,l}^{n,m}} \int_{P_{i,j}^{n,m}} |x-y|\, dydx
\end{aligned}$$

Now since x and y are in P_i^n, $|x-y| \leq \delta^n$, so:

$$\begin{aligned}
d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{\mu}_t^{n,m}, \overline{\nu}_t^{n,m}) &\leq n \sum_{i=1}^{n}\sum_{j=1}^{m}\sum_{l=1}^{m} n^{-2}m^{-2}\delta^n \\
&\leq \delta^n
\end{aligned}$$

By hypothesis $\delta^n \to 0$, it concludes. □

We have shown that we can make the empirical measure $\dot{\mu}^{n,m}$ arbitrary close to the averaged empirical measure $\dot{\nu}^{n,m}$ by taking large n but arbitrary m. Our next step is to remark the average empirical measure also satisfies the fixed point equation:

Lemma 3.4.11. *The averaged measure $\dot{\nu}^{n,m}$ satisfies the fixed point equation (FP) with averaged graphon and frequencies $\tilde{W}^n$, $\tilde{\omega}^n$ and initial data $\overline{\nu}_0^{n,m}$.*

Proof. Consider the velocity field $\mathcal{V}^{\dot{\nu}^{n,m}}$ from the equation of characteristics:

$$\mathcal{V}^{\dot{\nu}^{n,m}}(t,x,\Theta) = \begin{pmatrix} \phi \\ \tilde{\omega}^n(x) - D\phi + \int_\Omega \tilde{W}^n(x,y) \int_{\mathbb{T}\times\mathbb{R}} \sin(\xi-\theta-\beta)(\nu^{n,m})_t^y(d\xi, d\psi)dy \end{pmatrix}$$

The integral can be written:

$$\begin{aligned}\int_\Omega \tilde{W}^n(x,y)\int_{\mathbb{T}\times\mathbb{R}}\sin(\xi-\theta-\beta)(\nu^{n,m})^y_t(d\xi,d\psi)dy\\ &=\sum_{k=1}^n\int_{P^n_k}\frac{1}{m}\sum_{l=1}^m\tilde{W}^n(x,y)\sin(\tilde{\theta}^{n,m}_{k,l}(t)-\theta-\beta)dy\\ &=\frac{1}{nm}\sum_{k=1}^n\sum_{l=1}^m\tilde{W}^n(x,x^{n,m}_{k,l})\sin(\tilde{\theta}^{n,m}_{k,l}(t)-\theta-\beta)dy\end{aligned}$$

where we used that $\tilde{W}^n$ is constant on P^n_k. For any $x\in P^n_i$, we obtain exactly the velocity field acting on $\tilde{\theta}^{n,m}_{i,j}(t)$ for any j. We deduce that

$$\tilde{\Theta}^{n,m}_{i,j}(t)=T^{\dot{\nu}^{n,m}}_{t,0}(x,\tilde{\Theta}^{n,m}_{i,j}(0))$$

for any $x\in P^n_i$. This concludes since

$$T^{\dot{\nu}^{nm}}_{t,0}\sharp(\nu^{n,m})^x_0=\frac{1}{m}\sum_{j=1}^m\delta_{T^{\dot{\nu}^{nm}}_{t,0}(x,\tilde{\Theta}^{n,m}_{i,j}(0))}\text{ for }x\in P^n_i.$$

□

The next lemma is the most crucial one: it shows convergence of the averaged measure as $m\to\infty$ to a deterministic solution of the fixed point equation.

Lemma 3.4.12. *Let $\dot{\rho}^n$ be the solution of (FP) with averaged graphon and frequency $\tilde{W}^n$, $\tilde{\omega}^n$, and initial data $\bar{\rho}^n_0$ given by:*

$$(\rho^n)^x_0=n\big(\int_{P^n_i}\rho_0(x',\Theta)dx'\big)d\Theta,\quad \textit{for } x\in P^n_i.$$

Then, almost surely, for any $\alpha>c>0$, for any n,

$$d_\alpha(\dot{\nu}^{n,m},\dot{\rho}^n)\xrightarrow[m\to\infty]{}0.$$

Proof. We just showed that $\dot{\nu}^{n,m}$ also satisfied (FP) with same graphon and frequencies, so we only need to prove convergence of the initial data (by Lemma 3.3.7). We have

$$\begin{aligned}\bar{d}(\bar{\nu}^{n,m}_0,\bar{\rho}^n_0)&=\sum_{i=1}^n\int_{P^n_i}d\big((\nu^{n,m})^x_0,(\rho^n)^x_0\big)dx\\ &=\frac{1}{n}\sum_{i=1}^n d\big((\nu^{n,m})^{x^n_i}_0,(\rho^n)^{x^n_i}_0\big),\end{aligned}$$

since both measures are constant on P^n_i.

Now, we use that d metricizes the weak convergence of measures ([18], Theorem 11.3.3), so that it suffices to show:

$$\forall f\in BL(\mathbb{T}\times\mathbb{R}),\ \int_{\mathbb{T}\times\mathbb{R}}fd(\nu^{n,m})^{x^n_i}_0-\int_{\mathbb{T}\times\mathbb{R}}fd(\rho^n)^{x^n_i}_0\xrightarrow[m\to\infty]{}0.$$

Moreover, $BL(\mathbb{T}\times\mathbb{R})$ is separable (for the $\|\cdot\|_{L^\infty}$ norm, by Ascoli's Theorem). Considering $\{f_q\}_{q\in\mathbb{N}}$ a dense subspace, it thus suffices to show the convergence above for all f_q. We write, for any i, q:

$$\int_{\mathbb{T}\times\mathbb{R}} f_q d(\nu^{n,m})_0^{x_i^n} - \int_{\mathbb{T}\times\mathbb{R}} f_q d(\rho^n)_0^{x_i^n} = \frac{1}{m}\sum_{j=1}^m f_q(A_{i,j}^{n,m}) - n\int_{\mathbb{T}\times\mathbb{R}} f_q(\Theta)\int_{P_i^n}\rho_0(x,\Theta)dxd\Theta$$

where we recall that $A_{i,j}^{n,m}$ are the random variables giving the initial conditions, with law $\rho(x_{i,j}^{n,m},\cdot)$. The random variables $f_q(A_{i,j}^{n,m})$ have expectancy

$$\mathbb{E}f_q(A_{i,j}^{n,m}) = \int_{\mathbb{T}\times\mathbb{R}} f_q(\Theta)\rho_0(x_{i,j}^{n,m},\Theta)d\Theta$$

which yields, by correctly rearranging the terms,

$$\begin{aligned}
&\mathbb{E}\big(\frac{1}{m}\sum_{j=1}^m f_q(A_{i,j}^{n,m})\big) - n\int_{\mathbb{T}\times\mathbb{R}} f_q(\Theta)\int_{P_i^n}\rho_0(x,\Theta)dxd\Theta\\
&= \int_{\mathbb{T}\times\mathbb{R}} f_q(\Theta)\big(\big(\frac{1}{m}\sum_{j=1}^m \rho_0(x_{i,j}^{n,m},\Theta)\big) - n\int_{P_i^n}\rho_0(x,\Theta)dx\big)d\Theta\\
&= \int_{\mathbb{T}\times\mathbb{R}} f_q(\Theta)\frac{1}{m}\sum_{j=1}^m \big(\rho_0(x_{i,j}^{n,m},\Theta) - nm\int_{P_{i,j}^{n,m}}\rho_0(x,\Theta)dx\big)d\Theta\\
&= \int_{\mathbb{T}\times\mathbb{R}} f_q(\Theta)n\sum_{j=1}^m\int_{P_{i,j}^{n,m}}\big(\rho_0(x_{i,j}^{n,m},\Theta) - \rho_0(x,\Theta)\big)dxd\Theta.
\end{aligned}$$

We have $\left|x_{i,j}^{n,m} - x\right| \le \delta^{n,m} \xrightarrow[m\to\infty]{} 0$, so by using the uniform continuity of ρ_0 in x, we easily obtain that

$$\mathbb{E}\big(\frac{1}{m}\sum_{j=1}^m f_q(A_{i,j}^{n,m})\big) \xrightarrow[m\to\infty]{} n\int_{\mathbb{T}\times\mathbb{R}} f_q(\Theta)\int_{P_i^n}\rho_0(x,\Theta)dxd\Theta.$$

The random variables $f_q(A_{i,j}^{n,m})_j$ are independent and bounded. By the Strong Law of Large Numbers applied to the triangular array $(f_q(A_{i,j}^{n,m}))_{m\in\mathbb{N}^*,1\le j\le m}$, we get that

$$\frac{1}{m}\sum_{j=1}^m f_q(A_{i,j}^{n,m}) - \mathbb{E}\big(\frac{1}{m}\sum_{j=1}^m f_q(A_{i,j}^{n,m})\big) \xrightarrow[m\to\infty]{} 0$$

almost surely. Since $\{f_q\}_{q\in\mathbb{N}}$ is countable, almost surely this convergence holds for all f_q simultaneously. So at n an i fixed, almost surely $d\big((\nu^{n,m})_0^{x_i^n},(\rho^n)_0^{x_i^n}\big) \xrightarrow[m\to\infty]{} 0$. The set $\{n\in\mathbb{N}^*, 1\le i\le n\}$ is countable, so almost surely, for all n

$$\bar{d}(\bar{\nu}_0^{nm},\bar{\rho}_0^n) \xrightarrow[m\to\infty]{} 0.$$

Lemma 3.3.7 allows us to conclude convergence for all times. □

All is left now is to show convergence as $n\to\infty$ of $\dot{\rho}^n$ to $\dot{\mu}$.

Lemma 3.4.13. *Let $\epsilon > 0$, there exists n_3 such that, for $\alpha > c > 0$ and $n \geq n_3$, we have:*

$$d_\alpha(\dot{\rho}^n, \dot{\mu}) \leq \epsilon$$

Proof. We know that $\dot{\rho}^n$ solves (FP) with initial condition $\bar{\rho}_0^n$, graphon $\tilde{W}^n$, frequencies $\tilde{\omega}^n$, while $\dot{\mu}$ solves (FP) with initial condition $\bar{\mu}_0$, graphon W, frequencies ω. Using Lemmas 3.3.7, 3.3.8, 3.3.9, we only have to make the quantities $\left\|\tilde{W}^n - W\right\|_{L^1}$, $\|\tilde{\omega}^n - \omega\|_{L^1}$ and $\bar{d}(\bar{\rho}_0^n, \bar{\mu}_0)$ small. The former two go to zero thanks to Lemma 3.4.4. We detail the latter:

$$\bar{d}(\bar{\rho}_0^n, \bar{\mu}_0) = \int_\Omega d((\rho^n)_0^x, \mu_0^x) dx.$$

The measure $(\rho^n)_0^x$ is a probability with density given by

$$\rho^n(x, \Theta) = n \int_{P_i^n} \rho_0(y, \Theta) dy \text{ for } x \in P_i^n.$$

Let $i(n, x)$ be such that $x \in P_{i(n,x)}^n$, we have

$$|\rho^n(x, \Theta) - \rho_0(x, \Theta)| \leq n \int_{P_{i(n,x)}^n} |\rho_0(y, \Theta) - \rho_0(x, \Theta)| \, dy$$

Let $\varepsilon > 0$. Since x and y both are in $P_{i(n,x)}^n$, using that $|x - y| \leq \delta^n$ and the uniform continuity of ρ_0 in x, we have for n large enough, $|\rho_0(y, \Theta) - \rho_0(x, \Theta)| \leq \varepsilon$, which yields

$$|\rho^n(x, \Theta) - \rho_0(x, \Theta)| \leq n \int_{P_{i(n,x)}^n} \varepsilon dy = \varepsilon$$

Now we write:

$$\begin{aligned}
\bar{d}(\bar{\rho}_0^n, \bar{\mu}_0) &= \int_\Omega \sup_{f \in BL} \left| \int_{\mathbb{T} \times \mathbb{R}} f(\Theta) \big(\rho^n(x, \Theta) - \rho_0(x, \Theta)\big) d\Theta \right| dx \\
&\leq \int_\Omega \int_{\mathbb{T} \times \mathbb{R}} |\rho^n(x, \Theta) - \rho_0(x, \Theta)| \, d\Theta dx
\end{aligned}$$

We have $\rho^n(x, \Theta) \xrightarrow[n \to \infty]{} \rho_0(x, \Theta)$ pointwise, and

$$\int_\Omega \int_{\mathbb{T} \times \mathbb{R}} |\rho^n(x, \Theta)| \, d\Theta dx = 1 = \int_\Omega \int_{\mathbb{T} \times \mathbb{R}} |\rho_0(x, \Theta)| \, d\Theta dx$$

because $\rho_0 \geq 0$. By Scheffé's Lemma,

$$\int_\Omega \int_{\mathbb{T} \times \mathbb{R}} |\rho^n(x, \Theta) - \rho_0(x, \Theta)| \, d\Theta dx \xrightarrow[n \to \infty]{} 0.$$

This concludes. □

We end this section by proving Theorem 3.4.8.

Proof of Theorem 3.4.8. We use that for $\overline{\mu}$, $\overline{\nu}$ in $\bar{\mathcal{M}}$, the distance $d_{\Omega\times\mathbb{T}\times\mathbb{R}}$ is smaller than the distance $\bar{d}$:

$$\begin{aligned} d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{\mu},\overline{\nu}) &= \sup_{f\in BL(\Omega\times\mathbb{T}\times\mathbb{R})} \left| \int_\Omega \int_{\mathbb{T}\times\mathbb{R}} f(x,\Theta)(\mu^x(d\Theta)-\nu^x(d\Theta))dx \right| \\ &\leq \int_\Omega \sup_{f\in BL(\Omega\times\mathbb{T}\times\mathbb{R})} \left| \int_{\mathbb{T}\times\mathbb{R}} f(x,\Theta)(\mu^x(d\Theta)-\nu^x(d\Theta)) \right| dx \\ &\leq \int_\Omega d(\mu^x,\nu^x)dx \\ &= \bar{d}(\overline{\mu},\overline{\nu}) \end{aligned}$$

since $f(x,\cdot)\in BL(\mathbb{T}\times\mathbb{R})$. We fix $\alpha > c$ with c from the previous lemmas. Apply Lemmas **??**, 3.4.10 and 3.4.13 with $\epsilon = e^{-\alpha T}\varepsilon/4$ to get $n_0 = \max(n_1,n_2)$ such that, for all $n\geq n_0$, for all m, :

$$\begin{aligned} \sup_{t\in[0,T]} d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{(\mu^{n,m})}_t, \overline{(\nu^{n,m})}_t) &\leq e^{\alpha T} d_\alpha(\dot{\mu}^{n,m},\dot{\gamma}^{n,m}) \\ &\quad + \sup_{t\in[0,T]} d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{(\gamma^{n,m})}_t, \overline{(\nu^{n,m})}_t) \\ &\leq \varepsilon/2. \end{aligned}$$

We also have

$$\sup_{t\in[0,T]} d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{(\rho^n)}_t, \overline{\mu}_t) \leq \varepsilon/4,$$

by Lemma 3.4.13, up to changing n_0 to $\max(n_1,n_2,n_3)$.

Using Lemma **??**, for any fixed $n\geq n_0$, we have

$$d_\alpha(\dot{\nu}^{n,m},\dot{\rho}^n) \xrightarrow[m\to\infty]{} 0$$

almost surely. Choose m_0 such that

$$d_\alpha(\dot{\nu}^{n,m},\dot{\rho}^n) < e^{\alpha t}\varepsilon/4$$

for all $m\geq m_0$. We then have:

$$\sup_{t\in[0,T]} d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{(\nu^{n,m})}_t, \overline{(\rho^n)}_t) \leq \varepsilon/4.$$

Combine the three estimates to conclude. □

3.4.4 The case of randomly drawn intrinsic frequencies

We sketch how to adapt the analysis we conducted previously when the intrinsic frequencies of the Kuramoto Model are drawn randomly. We consider the Kuramoto model on N oscillators, where $(\omega_i^N)_{1\leq i\leq N}$ are independent random variable with probability distribution on $\mathbb{R}$ having density $g(\omega)$. We extend the phase space $\mathbb{T}\times\mathbb{R}$ to $\mathbb{T}\times\mathbb{R}\times\mathbb{R}$ and straightforwardly define the metric spaces $\mathcal{M}(\mathbb{T}\times\mathbb{R}\times\mathbb{R})$, $\bar{\mathcal{M}}(\mathbb{T}\times\mathbb{R}\times\mathbb{R})$ and $\dot{\mathcal{M}}_T(\mathbb{T}\times\mathbb{R}\times\mathbb{R})$, with their corresponding distances. The new empirical measures $\dot{\mu}^N$ are defined as:

$$(\mu^N)_t^x = \delta_{(\theta_i^N(t),\frac{d}{dt}\theta_i^N(t),w_i^N)} \text{ , for } x\in P_i^N.$$

The mean field limit for distribution of phases, velocities and frequencies $\mu_t^x(d\theta, d\phi, d\omega)$ is:

$$(\mathrm{MF}\omega) : \partial_t \mu_t^x + \phi\ \partial_\theta \mu_t^x + \partial_\phi\big(\mu_t^x\ V[\bar{\mu}_t](x,\theta,\phi,\omega)\big) = 0$$

with the acceleration field V given by:

$$V[\bar{\mu}_t](x,\theta,\phi,\omega) = \omega - D\phi + \int_\Omega W(x,y) \int_{\mathbb{T}\times\mathbb{R}} \sin(\xi - \theta - \beta)\mu(t,y,d\xi,d\psi,d\varpi)dy$$

We can conduct the same fixed point analysis, with the new velocity field in the equation of characteristics being:

$$\mathcal{V}^{\dot{\mu}}(t,x,\theta,\phi,\omega) = \begin{pmatrix} \phi \\ \omega - D\phi + \int_\Omega W(x,y) \int_{\mathbb{T}\times\mathbb{R}} \sin(\xi - \theta - \beta)\mu_t^y(d\xi,d\psi,d\varpi)dy \\ 0 \end{pmatrix}$$

Since the new component corresponding to ω is null, all estimates hold as before.

In the same framework as the previous section, and under the same hypothesis on the random initial data, graphon, and partition, we can follow the proof of Theorem 3.4.8 to show:

Theorem 3.4.14. *Consider $\dot{\mu}^{n,m}$ the empirical measure of (KMmn) with random initial conditions $(A_{i,j}^{n,m})_{i,j}$ and graphon $W^{n,m}$ as in the hypothesis of Theorem 3.4.8, and with frequencies $\omega_{i,j}^{n,m} = \omega_{in+j}^N$.*

Let $\dot{\mu}$ be the solution of (MFω) with initial data $\overline{\mu}_0$ given by

$$\mu_0^x = \rho_0(x,\Theta)g(\omega)d\Theta d\omega$$

and with graphon W. Then, almost surely, for any $\varepsilon > 0$, there exists n_0 such that, for any $n \geq n_0$, there exists m_0 so that for $m \geq m_0$:

$$\sup_{t\in[0,T]} d_{\Omega\times\mathbb{T}\times\mathbb{R}}(\overline{(\mu^{n,m})}_t, \overline{\mu}_t) < \varepsilon.$$

3.5 A brief introduction to graphons

In the former analysis, graphons have been used to model the limit behaviour of the sequence of adjacency matrices W^n, and the labelled cut distance $d_\square$ has been used to measure the dependence of the solution to the fixed point equation with respect to W. The goal of this section is to very broadly introduce an unfamiliar reader to the theory of graphons and motivate the distance $d_\square$. We refer to [24] for an in-depth review of the topic. Let us begin by giving the definition of graphons on a domain Ω.

Definition 3.5.1. *Let $\Omega \subset \mathbb{R}^d$ an open bounded set of Lebesgue measure 1. A graphon is a symmetric measurable function*

$$W : \Omega^2 \longrightarrow [0,1].$$

The space of graphons is $\mathcal{W} = \{W \in L^\infty(\Omega^2) \mid 0 \leq W \leq 1\ a.e.\}$

This brief and rather underwhelming definition requires some insight : Ω should be understood as an infinite set of vertices and $W(x,y)$ be the weight of the edge (x,y). For instance, given a weighted graph G (with normalized weight taking values in $[0,1]$), one can define an associated graphon :

Definition 3.5.2. *For any n, let $(P_i^n)_{1\leq i\leq n}$ be a partition of Ω in sets of equal Lebesgue measure $|P_i^n| = \frac{1}{n}$, which is fixed once and for all. Let G be a graph on n vertices with adjacency matrix $A = (a_{ij})$. The graphon associated to G, A and $(P_i)_{1\leq i\leq n}$ is defined by:*

$$W_G(x, y) = a_{ij}, \quad \forall (x, y) \in P_i \times P_j.$$

Note that this construction is not unique for a given graph G, not even after fixing the P_i^n, because any reordering of the labels of the vertices gives rise to a different adjacency matrix and thus a different graphon. This is something to keep in mind as we will define the notion of convergence of graphons.

Remark 3.5.3. *In the literature, Ω is simply $[0, 1]$, and $P_i^n = [\frac{i-1}{n}, \frac{i}{n}[$, which is sufficient to abstractly label the vertices of any graph of any size. However, since in our original problem, the graph associated to W^n describes a power grid, the vertices have a geographical location, being generators or machines. We choose to keep the general domain Ω for the vertices to represent the region of space the power system is located in.*

Convergence of graph sequences can then be described in terms of the convergence of their associated graphons. However, we cannot take any distance on the space $\mathcal{W}$, because we still want this distance to reflect the proximity of graphs. This role can be fulfilled by the cut distance $d_\square$. To briefly motivate its definition, consider two graphs G and H on the same set of vertices $\{1, 2, ..., n\}$, and their associated graphons W_G, W_H defined using the same partition $(P_i)_{1\leq i\leq n}$ of Ω. To compare H and G, we can pick two subsets of $\{1, 2, ..., n\}$, namely S' and T', and add the weight of edges between S' and T' in G and in H. Defining

$$S = \bigcup_{i\in S'} P_i \ , \quad T = \bigcup_{j\in T'} P_j \ ,$$

this can be expressed in terms of graphons by the following integral:

$$\int_S \int_T W_G(x, y) - W_H(x, y)\ dx\ dy.$$

If those quantities are close for any s and t, this should indicate that the two graphs themselves are close. This leads to the following definition:

Definition 3.5.4. *Let W, V be graphons. Their labelled cut distance is defined as*

$$d_\square(W, V) = \sup_{S,T\subset\Omega} \left| \int_S \int_T W(x, y) - V(x, y)\ dx\ dy \right|$$

where the supremum is taken over all measurable subsets of Ω.

The labelled cut distance is a distance on $\mathcal{W}$, which is strictly rougher than the L^1 norm. One property that will be useful later is that it is equivalent to the $L^\infty \to L^1$ operator norm in the following sense (see [47]):

Lemma 3.5.5. *There exists $C > 1$ such that, for all $W, V \in \mathcal{W}$,*

$$\frac{1}{C} d_\square(W, V) \ \leq \sup_{|f|\leq 1} \left\| \int_\Omega (W(\cdot, y) - V(\cdot, y)) f(y) dy \right\|_{L^1} \leq \ C d_\square(W, V).$$

As stated previously, the pairing between graphs and their graphons is far from unique. The cut distance between two graphons corresponding to the same graph but different labellings of the vertices has no reason to be zero or even small. In the applications we have in mind the vertices (machines or generators) have physical properties that distinguish them from another. This explains why we have mainly worked in the space of graphons $\mathcal{W}$ with its labelled cut distance $d_\square$. However, for general abstract graphs, labelling is arbitrary and the cut distance should be zero for such pairs of graphons. The natural way to remedy to this issue is to compare all possible choices of labels.

Definition 3.5.6. *Let W, V be graphons. We define their cut distance as*

$$\delta_\square(W,V) = \inf_\phi d_\square(W, V\circ\phi),$$

where the infimum is taken over all measure-preserving bijections ϕ of Ω, and $V\circ\phi(x,y) = V(\phi(x),\phi(y))$.

The cut distance between two graphons associated to the same graph is the same, provided there is a measure-preserving bijection sending P_i^n to $P_{\sigma(i)}^n$ for any permutation σ of the vertices, which we assume always exists (this is trivial in the case $\Omega = [0,1]$ and $P_i^n = [\frac{i-1}{n}, \frac{i}{n}[$, and its higher dimension counterparts) It remains now to identify two graphons at zero cut distance from each other, leading to the definition of the space of unlabelled graphons.

Definition 3.5.7. *The space of unlabelled graphons $\mathcal{W}_0$ is the quotient space of $\mathcal{W}$ under the equivalence relation*

$$W \sim V \iff \delta_\square(W,V) = 0$$

This is the natural space to study dense graph limits: indeed, it can be shown that $(\mathcal{W}_0, \delta_\square)$ is a compact metric space in which the set of (graphons associated to) finite graphs is dense [24].

4 Analysis on nearest-neighbour graphs

4.1 Notations

In this chapter we set aside the mean field approach and consider this time the Kuramoto model on sparser graphs, especially graph sequences where all vertices have finitely bounded degree. In the sense of graphons, such graphs converge to the null graphon, since the support of the associated graphon concentrates on a null Lebesgue set. In fact, even up to renormalization, the graphons we will consider do not have any meaningful limit in any L^p space, even endowed with weak topology. The mean field approach is thus of no use in this context.

Let us state our new framework and the main results of this chapter. We will work in two dimensions, which is the relevant case for practical applications. We once again consider n machines or generators, with positions $(x_i)_{1\le i\le n}$ in an open bounded set Ω of $\mathbb{R}^2$. We explicitly define $(P_i)_{1\le i\le n}$ to be the Voronoï partition associated to $(x_i)_{1\le i\le n}$, that is:

$$P_i = \{x\in\Omega \mid |x-x_i| < |x-x_j|,\ j\neq i\}.$$

Since n will be fixed in the proofs, we allowed ourselves to drop the explicit dependence on n. For the sake of simplicity, we assume Ω to be a polygonal convex subset of $\mathbb{R}^2$. This way, the P_i are polygonal and convex as well.

The oscillators at x_i and x_j are said to be neighbours (denoted by $i \sim j$) if P_i and P_j are adjacent, that is if they share an edge. We can write the edge ∂P_i of P_i as

$$\partial P_i = \bigcup_{j \sim i} \sigma_{ij}$$

where σ_{ij} is the line segment $\partial P_i \cap \partial P_j$. This union is disjoint up to null Lebesgue sets. We denote l_{ij} the length of the segment σ_{ij}, and $d_{ij} = |x_j - x_i|$.

4.2 The models

We consider the Kuramoto Model on the graph induced by the neighbourhood relation:

$$\text{(KMn)}: \ \frac{d^2}{dt^2}\theta_i(t) + D\frac{d}{dt}\theta_i(t) = \omega_i + \sum_{j=1}^{n} W_{ij}\sin(\theta_j(t) - \theta_i(t)).$$

This time we suppose $W_{ij} \neq 0$ if and only if $i \sim j$. Further hypothesis on the size of the coefficients will be made later. We fix a function ω and suppose the frequencies ω_i are given by

$$\omega_i = \frac{1}{|P_i|}\int_{P_i} \omega(x)dx.$$

The initial conditions are given by continuous functions θ_0, ϕ_0:

$$\theta_i(0) = \theta_0(x_i), \ \frac{d}{dt}\theta_i(0) = \phi_0(x_i).$$

Finally, D is assumed to be a positive coefficient.

Remark 4.2.1. *There is no phase shift in this model, $\beta = 0$. From a mathematical point of view, we need this to be able to linearize the sine when $\theta_j(t)$ and $\theta_i(t)$ are close. From a modelling point of view, recall that*

$$\beta = \arctan\left(\frac{x}{r}\right) - \frac{\pi}{2}$$

with r and x the real and imaginary part of the impedance per unit length z of the line. We are thus in the situation of $x \gg r$.

We define the piecewise constant function

$$\theta^n(t,x) = \sum_{j=1}^{n} \theta_j(t)\mathbf{1}_{P_j}(x).$$

Our goal is to show the convergence of θ^n to the solution θ of the continuous wave equation

$$\text{(W)}: \begin{cases} \partial_t^2\theta(t,x) + D\partial_t\theta(t,x) = \Delta\theta(t,x) + \omega(x) \ \text{ for all } (t,x) \in \mathbb{R}^+ \times \Omega, \\ \\ \nabla\theta(t,x)\cdot n(x) = 0 \text{ on } \partial\Omega, \ \theta(0,\cdot) = \theta_0, \ \partial_t\theta(0,\cdot) = \phi_0 \end{cases}$$

with Δ the Laplacian in the x variable and n the exterior normal to $\partial\Omega$.

The main result of this chapter is the following:

Theorem 4.2.2. *Suppose the following regularity: $\theta_0 \in H^4(\Omega)$, $\phi_0 \in H^3(\Omega)$, and $\omega \in H^2(\Omega)$. Fix n and let*

$$h = sup\{|x-y| \ | \ x,y \in P_i, \ 1 \leq i \leq n\}.$$

Suppose that the coefficients W_{ij} satisfy the following bound: Define

$$G(x) = \sum_{i=1}^{n} \left(\sum_{j \sim i} |W_{ij} - \frac{l_{ij}}{d_{ij} \, |P_i|}| \right) \mathbf{1}_{P_i}(x), \tag{4}$$

assume there exists C_G positive such that

$$\|G\|_{L^2} \leq C_G h. \tag{5}$$

Suppose the partition is regular enough: let $\zeta > 0$ be such that:

$$\forall i, \ \forall j, \ d_{ij} > \zeta h.$$

Let $\delta \in [\frac{8}{10}, 1[$, there exists h_0 depending on θ, C_G, Ω, D, T, δ and ζ, such that, if $h \leq h_0$, the following estimate holds for all $t \in [0,T]$:

$$\|\theta^n(t) - \theta(t)\|_{L^2} \leq C h^{\delta},$$

where C depends on Ω, θ and ζ.

If we have for each $n \in \mathbb{N}^*$, a sequence of points $(x_i^n)_{1 \leq i \leq n}$, and coefficients W_{ij}^n for which the partition $(P_i^n)_{1 \leq i \leq n}$ satisfies the hypothesis of Theorem 4.2.2 with constants C_G and ζ uniform in n, we thus have convergence of the finite Kuramoto model to the wave equation.

Remark 4.2.3. *The proof relies on viewing (KMn) as a (non-linear) finite volume scheme for equation (W), following the methods of [20]. This is why the W_{ij} must converge to the coefficients of the natural finite volume approximation of (W). To allow for a wider range of coefficients W_{ij}, one can consider more general elliptic operators of the type $div(\Lambda(x)\nabla \cdot)$, leading to different limit coefficients in the finite volume scheme.*

However, for such operators, the partition P_i must be adapted, as it is the case in numerical analysis. The so-called admissible meshes are described in [20], definition 3.7. As it is quite lengthy and purely technical, but the proofs are essentially the same, we omit to elaborate further on this definition. The coefficients themselves have a convoluted expression that is of little interest in the present discussion.

The proof of Theorem 4.2.2 is divided in three steps : we first write a finite volume scheme for equation (W), and see how it is satisfied up to an error by both θ^n and θ. We then establish a discrete energy estimate for the difference ε between θ^n and θ. Finally, we control each of the error terms in the energy estimate to obtain a bound on ε which goes to 0 as $n \to \infty$. Before the bulk of the proof, we recall some results on the existence, uniqueness and regularity of the solution θ of (W).

4.3 The solution of the wave equation

Here we recall some results on the regularity of the solutions of wave equations. We can proceed to some reductions: up to replacing θ by

$$\theta(t,x) - \left(\frac{t}{D} \int_\Omega \omega(y)dy + \frac{e^{-Dt}-1}{D} \int_\Omega \phi_0(y) - \omega(y)dy + \int_\Omega \theta_0(y)dy \right),$$

and doing the same with θ^n, we can assume that every function in play has zero mean value:

$$\int_\Omega \omega(y)dy = \int_\Omega \phi_0(y)dy = \int_\Omega \theta_0(y)dy = 0.$$

Usual semigroup theory on the wave equation with Neumann boundary conditions ensures that, provided an initial data $(\theta_0, \phi_0) \in H^4(\Omega) \times H^3(\Omega)$, there exists a unique solution θ of (W), that belongs to the following space:

$$\theta \in C^2([0,T]; H^2(\Omega)) \cap C([0,T]; H^4(\Omega)).$$

Remark that since $H^2(\Omega)$ is embedded into $C(\bar{\Omega})$, we have in addition

$$\theta \in C^2([0,T] \times \bar{\Omega})$$

and all functions appearing in (W) are continuous, so that the equation is satisfied for all x and not only almost everywhere.

4.4 The finite volume scheme

Here we establish a finite volume scheme for the wave equation following the usual technique: we integrate (W) over P_i to obtain:

$$\int_{P_i} \partial_t^2 \theta(t,x)dx + D \int_{P_i} \partial_t \theta(t,x)dx = \int_{P_i} \Delta\theta(t,x)dx + \int_{P_i} \omega(x)dx.$$

We use the Stokes formula on the first term of the right hand side to rewrite it as:

$$\begin{aligned} \int_{P_i} \Delta\theta(t,x)dx &= \int_{\partial P_i} \nabla\theta(t,x) \cdot n_i(x)\ dl(x) \\ &= \sum_{j \sim i} \int_{\sigma_{ij}} \nabla\theta(t,x) \cdot n_{ij}\ dl(x) \end{aligned}$$

where n_i is the exterior normal to ∂P_i^n, constant on σ_{ij} equal to n_{ij} ; and dl the line measure. Since P_i^n is a Voronoï partition, this ensures that n_{ij} is parallel to the line segment $[x_i, x_j]$. We approximate the integral by a discrete gradient:

$$\int_{\sigma_{ij}} \nabla\theta(t,x) \cdot n_{ij}\ dl(x) = l_{ij} \frac{\theta(t,x_j) - \theta(t,x_i)}{d_{ij}} + l_{ij} R_{ij}(t)$$

where $R_{ij}(t)$ is the discretization error

$$R_{ij}(t) = \frac{1}{l_{ij}} \int_{\sigma_{ij}} \nabla\theta(t,x) \cdot n_{ij}\ dl(x) - \frac{\theta(t,x_j) - \theta(t,x_i)}{d_{ij}}.$$

We also define M_i the error between the values of the derivatives of θ and their mean value:

$$M_i(t) = \partial_t^2\theta(t,x_i) + D\partial_t\theta(t,x_i) - \frac{1}{|P_i|}\int_{P_i}\partial_t^2\theta(t,x)dx - \frac{D}{|P_i|}\int_{P_i}\partial_t\theta(t,x)dx.$$

We can thus write that θ satisfies a finite volume scheme up to errors M_i and R_{ij}:

$$\partial_t^2\theta(t,x_i) + D\partial_t\theta(t,x_i) = \omega_i + \sum_{j\sim i}\frac{l_{ij}}{d_{ij}\,|P_i|}\big(\theta(t,x_j) - \theta(t,x_i)\big) + M_i(t) + \sum_{j\sim i}\frac{l_{ij}}{|P_i|}R_{ij}(t). \quad (6)$$

This is a linearized Kuramoto model with coefficients $V_{ij} = \frac{l_{ij}}{d_{ij}|P_i|}$. We can thus expect the finite model to satisfy the scheme up to a small error:

$$\frac{d^2}{dt^2}\theta_i(t) + D\frac{d}{dt}\theta_i(t) = \omega_i + \sum_{j\sim i}V_{ij}(\theta_j(t)-\theta_i(t)) + \sum_{j\sim i}\frac{l_{ij}}{|P_i|}S_{ij}(t) + \sum_{j\sim i}G_{ij}\sin(\theta_j(t)-\theta_i(t)). \quad (7)$$

There are two differences between the Kuramoto model and the finite volume scheme: the linearization of the sine entails an error S_{ij}:

$$S_{ij} = \frac{\sin(\theta_j(t)-\theta_i(t)) - (\theta_j(t)-\theta_i(t))}{d_{ij}},$$

and replacing the coefficients W_{ij} by V_{ij} leads to the error term G_{ij}:

$$G_{ij} = W_{ij} - V_{ij}.$$

The goal is now to exploit those identities to control the difference between θ and θ^n.

4.5 The energy estimate

By linearity, the difference $\varepsilon_i(t) = \theta_i(t) - \theta(t,x_i)$ satisfies the scheme, up to all the errors. We can use this to derive a discrete version of the usual energy estimate for solutions of the wave equation. We let

$$\varepsilon(t,x) = \sum_{j=1}^{n}\varepsilon_j(t)\mathbf{1}_{P_j}(x).$$

For quantities depending on the cells P_i, such as the mean error M_i, we define the piecewise constant function on each P_i:

$$M(t,x) = \sum_{j=1}^{n}M_j(t)\mathbf{1}_{P_j}(x),$$

For quantities depending on the edges σ_{ij}, it is more natural to use different cells to build step functions. We thus define the triangular shaped cells T_{ij}, which have for vertices x_i and the two ends of the segment σ_{ij}. The triangles $\{T_{ij}\}_{ij}$ form a partition of Ω, and we have

$$P_i = \bigcup_{j\sim i}T_{ij}.$$

In addition, we define the diamond shaped cells $\langle ij \rangle = T_{ij} \cup T_{ji}$, which have (perpendicular) diagonals $[x_i, x_j]$ and σ_{ij}. To sum over the edges, we use the following notation:

$$\sum_{\sigma_{ij}} = \sum_{\substack{i,j \\ i<j, i\sim j}}$$

This allows us to define the discrete gradient of ε:

$$\nabla_d \varepsilon(t,x) = \sum_{\sigma_{ij}} \frac{\varepsilon_j - \varepsilon_i}{d_{ij}} \mathbf{1}_{\langle ij \rangle}(x),$$

as well as the error functions

$$R(t,x) = \sum_{\sigma_{ij}} R_{ij}(t) \mathbf{1}_{\langle ij \rangle}(x),$$

and

$$S(t,x) = \sum_{\sigma_{ij}} \frac{S_{ij}(t)}{d_{ij}} \mathbf{1}_{\langle ij \rangle}(x).$$

The discrete energy estimate gives a control over the difference ε in terms of the error functions:

Lemma 4.5.1. *The function ε satisfies the following energy estimate, where C depends only on D and T:*

$$\begin{aligned} \|\partial_t \varepsilon(t)\|_{L^2}^2 + \|\nabla_d \varepsilon(t)\|_{L^2}^2 + \frac{D}{2} \int_0^t \|\partial_t \varepsilon(s)\|_{L^2}^2 \, ds \leq C \int_0^t \|\partial_t R(s)\|_{L^2}^2 \, ds + C \|R(t)\|_{L^2}^2 \\ + C \int_0^t \|S(s)\|_{L^2}^2 \, ds + C \int_0^t \|M(s)\|_{L^2}^2 \, ds \\ + CT \|G\|_{L^2}^2 . \end{aligned}$$

Proof. We write the system verified by ε_i, obtained by subtracting the identity (6) for $\theta(t, x_i)$ to the equation (7) for θ_i:

$$\begin{aligned} \frac{d^2}{dt^2}\varepsilon_i(t) + D\frac{d}{dt}\varepsilon_i(t) = &\sum_{j\sim i} V_{ij}(\varepsilon_j(t) - \varepsilon_i(t)) \\ &+ \sum_{j\sim i} \frac{l_{ij}}{|P_i|}(S_{ij}(t) - R_{ij}(t)) + \sum_{j\sim i} G_{ij} \sin(\theta_j(t) - \theta_i(t)) - M_i(t). \end{aligned}$$

In the continuous case, one would multiply by $\partial_t \varepsilon$ and integrate over Ω. Here we multiply by $\frac{d}{dt}\varepsilon_i$ and sum over i to discover:

$$\begin{aligned} \sum_{i=1}^n |P_i| \left(\frac{d^2}{dt^2}\varepsilon_i \frac{d}{dt}\varepsilon_i + D(\frac{d}{dt}\varepsilon_i)^2 \right) = &\sum_{i=1}^n \frac{d}{dt}\varepsilon_i \sum_{j\sim i} \frac{l_{ij}}{d_{ij}}(\varepsilon_j - \varepsilon_i) + \sum_{i=1}^n \frac{d}{dt}\varepsilon_i \sum_{j\sim i} l_{ij}(S_{ij} - R_{ij}) \\ &+ \sum_{i=1}^n \frac{d}{dt}\varepsilon_i \sum_{j\sim i} |P_i| \, G_{ij} \sin(\theta_j - \theta_i) - \sum_{i=1}^n \frac{d}{dt}\varepsilon_i \, |P_i| \, M_i \end{aligned}$$

Note that we dropped the argument t for compactness. We now proceed to discrete integration by parts. Our goal is now to rewrite every term as integrals over Ω. The simple sums can directly be rewritten, using that the P_i form a partition and that the functions in play are piecewise constant:

$$\sum_{i=1}^{n} |P_i| \left(\frac{d^2}{dt^2}\varepsilon_i \frac{d}{dt}\varepsilon_i + D(\frac{d}{dt}\varepsilon_i)^2 \right) = \frac{1}{2}\frac{d}{dt}\int_\Omega (\partial_t \varepsilon(t,x))^2 dx + D\int_\Omega (\partial_t \varepsilon(t,x))^2 dx$$

We have as well:

$$\sum_{i=1}^{n} \frac{d}{dt}\varepsilon_i \, |P_i| \, M_i = \int_\Omega M(t,x)\partial_t \varepsilon(t,x) dx.$$

We now turn to the other more complicated terms. The first double sum can be rewritten as a sum over all edges σ_{ij}. Each edge σ_{ij} appears exactly twice, once with the term $\frac{d}{dt}\varepsilon_i \frac{l_{ij}}{d_{ij}}(\varepsilon_j - \varepsilon_i)$ and once with $\frac{d}{dt}\varepsilon_j \frac{l_{ij}}{d_{ij}}(\varepsilon_i - \varepsilon_j)$. We can thus write

$$\sum_{i=1}^{n} \frac{d}{dt}\varepsilon_i \sum_{j\sim i} \frac{l_{ij}}{d_{ij}}(\varepsilon_j - \varepsilon_i) = \sum_{\sigma_{ij}} \frac{l_{ij}}{d_{ij}}(\varepsilon_j - \varepsilon_i)\frac{d}{dt}(\varepsilon_i - \varepsilon_j).$$

The sum then becomes

$$\begin{aligned}\sum_{\sigma_{ij}} \frac{l_{ij}}{d_{ij}}(\varepsilon_j - \varepsilon_i)\frac{d}{dt}(\varepsilon_i - \varepsilon_j) &= -\frac{1}{2}\frac{d}{dt}\sum_{\sigma_{ij}} \frac{l_{ij}}{d_{ij}}(\varepsilon_j - \varepsilon_i)^2 \\ &= -\frac{1}{2}\frac{d}{dt}\sum_{\sigma_{ij}} l_{ij} d_{ij} \left(\frac{\varepsilon_j - \varepsilon_i}{d_{ij}}\right)^2\end{aligned}$$

Remark that the diamond-shaped cell has an area $|\langle ij \rangle| = \frac{1}{2} d_{ij} l_{ij}$, so

$$-\frac{1}{2}\frac{d}{dt}\sum_{\sigma_{ij}} l_{ij} d_{ij} \left(\frac{\varepsilon_j - \varepsilon_i}{d_{ij}}\right)^2 = -\frac{d}{dt}\int_\Omega (\nabla_d \varepsilon(t,x))^2 dx$$

We can apply the same operation to the next double sum, using the antisymmetry of the terms R_{ij} and S_{ij} to obtain:

$$\begin{aligned}\sum_{i=1}^{n} \frac{d}{dt}\varepsilon_i \sum_{j\sim i} l_{ij}(S_{ij} - R_{ij}) &= \sum_{\sigma_{ij}} l_{ij}(S_{ij} - R_{ij})\frac{d}{dt}(\varepsilon_i - \varepsilon_j) \\ &= -\int_\Omega \big(\tilde{S}(t,x) - R(t,x)\big)\partial_t \nabla_d \varepsilon(t,x) dx.\end{aligned}$$

by dividing and multiplying by d_{ij}, and defining $\tilde{S}$ like S but without the d_{ij}:

$$\tilde{S}(t,x) = \sum_{\sigma_{ij}} S_{ij}(t) \mathbf{1}_{\langle ij \rangle}(x).$$

Finally, we keep the last double sum as it is, since we lack antisymmetry for G_{ij}. We will bound it later, for now we let:

$$\mathcal{G}(t) = \sum_{i=1}^{n} \frac{d}{dt}\varepsilon_i \sum_{j\sim i} |P_i|\, G_{ij} \sin(\theta_j - \theta_i).$$

Combining everything, we obtain the identity, holding for all t:

$$\frac{1}{2}\frac{d}{dt}\int_\Omega (\partial_t\varepsilon(t,x))^2 dx + D\int_\Omega (\partial_t\varepsilon(t,x))^2 dx + \frac{d}{dt}\int_\Omega (\nabla_d\varepsilon(t,x))^2 dx = \\ -\int_\Omega \big(\tilde{S}(t,x) - R(t,x)\big)\partial_t\nabla_d\varepsilon(t,x)dx - \int_\Omega M(t,x)\partial_t\varepsilon(t,x)dx + \mathcal{G}(t).$$

Since $\varepsilon(0,x) = \nabla_d\varepsilon(0,x) = \partial_t\varepsilon(0,x) = 0$ thanks to the initial conditions, integrating over $[0,t]$ yields:

$$\frac{1}{2}\int_\Omega (\partial_t\varepsilon(t,x))^2 dx + \int_\Omega (\nabla_d\varepsilon(t,x))^2 dx + D\int_0^t\int_\Omega (\partial_t\varepsilon(s,x))^2 dxds = \\ + \int_0^t\int_\Omega R(s,x)\partial_t\nabla_d\varepsilon(s,x)dxds - \int_0^t\int_\Omega \tilde{S}(s,x)\partial_t\nabla_d\varepsilon(s,x)dxds \\ - \int_0^t\int_\Omega M(s,x)\partial_t\varepsilon(s,x)dxds + \int_0^t \mathcal{G}(s)ds.$$

We are close to the desired result, we simply have to bound the right hand side so the error terms appear more explicitly.

• *The R error term:* We perform integration by parts in time on this first term:

$$\int_0^t\int_\Omega R(s,x)\partial_t\nabla_d\varepsilon(s,x)dxds = -\int_0^t\int_\Omega \partial_t R(s,x)\nabla_d\varepsilon(s,x)dxds + \int_\Omega R(t,x)\nabla_d\varepsilon(t,x)dx,$$

so we can bound it, using Cauchy-Schwarz and Young inequalities:

$$\left|\int_0^t\int_\Omega R(s,x)\partial_t\nabla_d\varepsilon(s,x)dxds\right| \le \frac{1}{2}\int_0^t\int_\Omega (\partial_t R(s,x))^2 dsdx + \frac{1}{2}\int_0^t\int_\Omega (\nabla_d\varepsilon(s,x))^2 dsdx \\ + \frac{1}{2}\int_\Omega (R(t,x))^2 dx + \frac{1}{2}\int_\Omega (\nabla_d\varepsilon(t,x))^2 dx.$$

• *The $\tilde{S}$ error term:* In this term, it is not pertinent to integrate by parts in time, as the derivative of S_{ij} is harder to control.

$$\left|\int_\Omega \tilde{S}(s,x)\partial_t\nabla_d\varepsilon(s,x)dx\right| = \left|\sum_{\sigma_{ij}} l_{ij}S_{ij}\frac{d}{dt}(\varepsilon_i - \varepsilon_j)\right| \\ \le 4\sum_{\sigma_{ij}} \frac{1}{4}l_{ij}d_{ij}\left|\frac{S_{ij}}{d_{ij}}\right|\left(\left|\frac{d\varepsilon_i}{dt}\right| + \left|\frac{d\varepsilon_j}{dt}\right|\right) \\ \le$$

Notice that the product of functions $|S|\,|\partial_t\varepsilon|$ is equal to $\left|\frac{S_{ij}}{d_{ij}}\right|\left|\frac{d\varepsilon_i}{dt}\right|$ on the triangle T_{ij}, which has area $\frac{1}{4}l_{ij}d_{ij}$, so we can write:

$$4\sum_{\sigma_{ij}}\frac{1}{4}l_{ij}d_{ij}\left|\frac{S_{ij}}{d_{ij}}\right|\left(\left|\frac{d\varepsilon_i}{dt}\right|+\left|\frac{d\varepsilon_j}{dt}\right|\right)=4\int_\Omega|S(s,x)|\,|\partial_t\varepsilon(s,x)|\,dx.$$

With a constant C depending on D, we can write:

$$4\int_\Omega|S(s,x)|\,|\partial_t\varepsilon(s,x)|\,dx\leq C\int_\Omega(S(s,x))^2dx+\frac{D}{4}\int_\Omega(\partial_t\varepsilon(s,x))^2dx$$

The final control on this term thus is, after integration in time:

$$\left|\int_0^t\int_\Omega\tilde{S}(s,x)\partial_t\nabla_d\varepsilon(s,x)dxdx\right|\leq C\int_0^t\int_\Omega(S(s,x))^2dx+\frac{D}{4}\int_0^t\int_\Omega(\partial_t\varepsilon(s,x))^2dx.$$

- *The M error term:* We can deal with this term using Cauchy-Schwarz and Young inequalities again:

$$\left|\int_0^t\int_\Omega M(s,x)\partial_t\varepsilon(s,x)dxds\right|\leq C\int_0^t\int_\Omega(M(s,x))^2dxds+\frac{D}{4}\int_0^t\int_\Omega(\partial_t\varepsilon(s,x))^2dxds$$

where we possibly change C, but still depending on D only.

- *The G error term:* Finally, we write that

$$\begin{aligned}|\mathcal{G}(s)|&=\left|\sum_{i=1}^n\frac{d\varepsilon_i}{dt}\left(\sum_{j\sim i}|P_i|\,G_{ij}\sin(\theta_j-\theta_i)\right)\right|\\&\leq\sum_{i=1}^n|P_i|\left|\frac{d\varepsilon_i}{dt}\right|\left(\sum_{j\sim i}|G_{ij}|\right)\end{aligned}$$

Using the definition of G (4) from the hypothesis of Theorem 4.2.2, we have:

$$\sum_{i=1}^n|P_i|\left|\frac{d\varepsilon_i}{dt}\right|\left(\sum_{j\sim i}|G_{ij}|\right)\leq C\int_\Omega(G(x))^2dx+\frac{D}{4}\int_\Omega(\partial_t\varepsilon(s,x))^2dx$$

using usual inequalities, and the final bound after integration in time is:

$$\left|\int_0^t\mathcal{G}(s)ds\right|\leq TC\int_\Omega(G(x))^2dx+\frac{D}{4}\int_0^t\int_\Omega(\partial_t\varepsilon(s,x))^2dxds.$$

Combining all these bounds, we obtain an estimate:

$$\begin{aligned}\frac{1}{2}\int_\Omega(\partial_t\varepsilon(t,x))^2dx+\frac{1}{2}\int_\Omega(\nabla_d\varepsilon(t,x))^2dx+\frac{D}{4}\int_0^t\int_\Omega(\partial_t\varepsilon(s,x))^2dxds\\ \leq\frac{1}{2}\int_0^t\int_\Omega(\nabla_d\varepsilon(s,x))^2dxds\\ +\frac{1}{2}\int_0^t\int_\Omega(\partial_tR(s,x))^2dxds+\frac{1}{2}\int_\Omega(R(t,x))^2dx\\ +C\int_0^t\int_\Omega(S(s,x))^2dxds+C\int_0^t\int_\Omega(M(s,x))^2dxds\\ +TC\int_\Omega(G(x))^2dx.\end{aligned}$$

We can now apply Gronwall's Lemma (3.3.2) to obtain

$$\begin{aligned}\|\partial_t\varepsilon(t)\|_{L^2}^2+\|\nabla_d\varepsilon(t)\|_{L^2}^2+\frac{D}{2}\int_0^t\|\partial_t\varepsilon(s)\|_{L^2}^2\,ds\leq C\int_0^t\|\partial_tR(s)\|_{L^2}^2\,ds+C\|R(t)\|_{L^2}^2\\ +C\int_0^t\|S(s)\|_{L^2}^2\,ds+C\int_0^t\|M(s)\|_{L^2}^2\,ds\\ +CT\|G\|_{L^2}^2.\end{aligned}$$

with a constant C now depending on D and T, which is the desired result. □

To obtain Theorem 4.2.2, we now need to show each error is controlled by the grid size h.

4.6 Controlling the errors

4.6.1 The discretization errors

The error terms in R are due to the approximation of the flux by a discrete gradient. The next lemma, adapted from [20], gives consistency of this approximation:

Lemma 4.6.1. *There exists C_R depending on ζ and θ such that, for any $s\in[0,T]$:*

$$\|R(s)\|_{L^2}^2\leq C_Rh^2,$$

$$\|\partial_tR(s)\|_{L^2}^2\leq C_Rh^2.$$

Proof. We begin by showing the first estimate. Fix an edge σ_{ij}, recall that

$$R_{ij}(s)=\frac{1}{l_{ij}}\int_{\sigma_{ij}}\nabla\theta(s,x)\cdot n_{ij}\;dl(x)-\frac{\theta(s,x_j)-\theta(s,x_i)}{d_{ij}}.$$

Let $x_m=\frac{1}{2}(x_i+x_j)$ be the middle of the segment $[x_i,x_j]$. It verifies $x_m\in\sigma_{ij}$ since the partition is a Voronoï mesh. Recall that $\theta\in C^2([0,T]\times\bar{\Omega})$), so we can easily perform Taylor expansions: there exists C_θ such that

$$\theta(s,x_j)=\theta(s,x_m)+\nabla\theta(s,x_m)\cdot(x_j-x_m)+\epsilon_i$$

$$\theta(s,x_i) = \theta(s,x_m) + \nabla\theta(s,x_m)\cdot(x_i - x_m) + \epsilon_j$$

where $|\epsilon_{i,j}| \leq C_\theta |x_{i,j} - x_m|^2$. By subtracting one line from the other and dividing by $d_{ij} = |x_i - x_j|$, we deduce:

$$\left| \frac{\theta(s,x_j) - \theta(s,x_i)}{d_{ij}} - \nabla\theta(s,x_m)\cdot\frac{(x_j - x_i)}{d_{ij}} \right| \leq C_\theta d_{ij}$$

with a possible change of C_θ. Since σ_{ij} and $[x_i, x_j]$ are orthogonal, the normal is given by

$$n_{ij} = \frac{(x_j - x_i)}{d_{ij}}.$$

Now remark that, for any $x \in \sigma_{ij}$, we also have by C^1 regularity of the gradient:

$$|\nabla\theta(s,x)\cdot n_{ij} - \nabla\theta(s,x_m)\cdot n_{ij}| \leq C_\theta |x - x_m|$$

Using that both $x - x_m$ and d_{ij} are smaller than h, we get:

$$|R_{ij}(s)| \leq C_\theta h.$$

By definition of R,

$$\begin{aligned}
\|R(s)\|_{L^2}^2 &= \sum_{\sigma_{ij}} |\langle ij\rangle| \, (R_{ij}(s))^2 \\
&\leq \sum_{\sigma_{ij}} |\langle ij\rangle| \, C_\theta^2 h^2 \\
&\leq C_\theta^2 h^2
\end{aligned}$$

Letting $C_R = C_\theta^2$ gives the first estimate. For the second one, we lack the C^2 regularity. Indeed,

$$\frac{d}{dt} R_{ij}(s) = \frac{1}{l_{ij}} \int_{\sigma_{ij}} \nabla\partial_t\theta(s,x)\cdot n_{ij} \; dl(x) - \frac{\partial_t\theta(s,x_j) - \partial_t\theta(s,x_i)}{d_{ij}}.$$

which is exactly $R_{ij}(s)$ but with θ replaced by $\partial_t\theta$. We only have $\partial_t\theta(s) \in H^3(\Omega)$, so Sobolev embeddings do not yield C^2 regularity. However, it is possible to adapt the previous proof using Taylor expansions with integral reminder instead of the blunt bounds we used. This is done in [20], in the proof of Theorem 3.1.6, and is purely technical so we will do not detail it here. The following inequality holds as soon as $\partial_t\theta(s) \in H^2(\Omega)$, which is the case here:

$$|\frac{d}{dt} R_{ij}(s)|^2 \leq C \frac{h^2}{l_{ij} d_{ij}} \int_{\langle ij\rangle} |H(\partial_t\theta(s))|^2(x) dx,$$

where $H(\partial_t\theta(s))$ is the Hessian matrix of $\partial_t\theta(s)$, and C depends this time on ζ. Plugging this in the expression of the L^2 norm yields:

$$\begin{aligned}
\|\partial_t R(s)\|_{L^2}^2 &\leq Ch^2 \sum_{\sigma_{ij}} \frac{|\langle ij\rangle|}{l_{ij} d_{ij}} \int_{\langle ij\rangle} |H(\partial_t\theta(s))|^2(x) dx \\
&\leq Ch^2 \sum_{\sigma_{ij}} \int_{\langle ij\rangle} |H(\partial_t\theta(s))|^2(x) dx \\
&\leq Ch^2 \int_\Omega |H(\partial_t\theta(s))|^2(x) dx \\
&\leq Ch^2 \, \|\partial_t\theta(s)\|_{H^2}^2 \,,
\end{aligned}$$

where we first used that $|\langle ij\rangle| = \frac{1}{2} l_{ij} d_{ij}$, and then that the $\langle ij\rangle$ are disjoint and the integrand is positive. This concludes the proof. □

The error terms in M are due to replacing the value of the functions at x_i by their mean value over P_i. With enough regularity we can have explicit bounds on these errors as well.

Lemma 4.6.2. *There exists C_M depending on θ such that, for any $s \in [0,T]$:*

$$\|M(s)\|_{L^2}^2 \leq C_M h^2.$$

Proof. Consider an arbitrary function $u \in C^2(\bar{\Omega})$. The difference between its value at x_i and its mean value over P_i is

$$\begin{aligned}\mathcal{M}_i = \frac{1}{|P_i|}\int_{P_i} u(x)dx - u(x_i) &= \frac{1}{|P_i|}\int_{P_i}(u(x)-u(x_i))dx \\ &= \frac{1}{|P_i|}\int_{P_i}\int_0^1 \nabla u(x_i + t(x-x_i))\cdot(x-x_i)dtdx\end{aligned}$$

by the fundamental theorem of analysis. Using a change of variable $z = x_i + t(x-x_i)$, we have:

$$\mathcal{M}_i = \frac{1}{|P_i|}\int_0^1 \int_{tP_i} \nabla u(z)\cdot(z-x_i)t^{-d-1}dzdt$$

We then apply Holder inequality, with $p=6$, $q=\frac{6}{5}$:

$$\begin{aligned}\mathcal{M}_i &\leq \frac{1}{|P_i|}\left(\int_0^1\int_{tP_i}|\nabla u(z)|^p\,dzdt\right)^{\frac{1}{p}}\left(\int_0^1\int_{tP_i}|z-x_i|^q t^{-dq-q}dzdt\right)^{\frac{1}{q}} \\ &\leq \frac{1}{|P_i|}\left(\int_{P_i}|\nabla u(z)|^p\,dz\right)^{\frac{1}{p}}\left(\int_0^1\int_{tP_i}h^q t^{-dq}dzdt\right)^{\frac{1}{q}}\end{aligned}$$

bounding the integral over tP_i by the one over the whole P_i on the first factor, and using that $|z-x_i| \leq th$ for the second factor. Continuing through, one gets, since $|tP_i| = t^d\,|P_i|$:

$$\begin{aligned}\mathcal{M}_i &\leq \frac{h}{|P_i|}\left(\int_{P_i}|\nabla u(z)|^p\,dz\right)^{\frac{1}{p}}\left(\int_0^1 |P_i|\,t^{d-dq}dt\right)^{\frac{1}{q}} \\ &= \frac{h}{|P_i|^{\frac{1}{p}}}\left(\int_{P_i}|\nabla u(z)|^p\,dz\right)^{\frac{1}{p}}\left(\int_0^1 t^{d-dq}dt\right)^{\frac{1}{q}} \\ &= C\frac{h}{|P_i|^{\frac{1}{p}}}\left(\int_{P_i}|\nabla u(z)|^p\,dz\right)^{\frac{1}{p}}\end{aligned}$$

Indeed, $d-dq = -\frac{2}{5} > -1$ and the integral in the second factor is finite. We now estimate the

L^2 norm, which since Ω is bounded is controlled by the L^p norm thanks to Holder inequality:

$$\begin{aligned}\sum_{i=1}^{n} |P_i| \, |\mathcal{M}_i|^2 &\leq \left(\sum_{i=1}^{n} |P_i| \, |\mathcal{M}_i|^p \right)^{\frac{2}{p}} \\ &\leq Ch^2 \left(\sum_{i=1}^{n} \int_{P_i} |\nabla u(z)|^p \, dz \right)^{\frac{2}{p}} \\ &\leq Ch^2 \, \|\nabla u\|_{L^p}^2 \, .\end{aligned}$$

Thanks to the Sobolev embedding $H^1 \hookrightarrow L^p$, we have:

$$\sum_{i=1}^{n} |P_i| \, |\mathcal{M}_i|^2 \leq Ch^2 \, \|u\|_{H^2}^2 \, .$$

By a density argument using C^2 functions, this holds for any $u \in H^2(\Omega)$ (which is always continuous so $u(x_i)$ is defined). We apply it with $u = \partial_t \theta(s)$ and $u = \partial_t^2 \theta(s)$ to obtain the desired result. □

4.6.2 The linearization error

It remains to bound the error S due to the linearization of the sine, which is the most complex term to deal with. We first establish an auxiliary estimate:

Lemma 4.6.3. *There exists C a constant depending only on Ω and ζ, such that, for any $K > 0$, for any $s \in [0, T]$, the following estimate holds:*

$$\|S(s)\|_{L^2}^2 \leq Ch^2 \, \|\theta(s)\|_{H^4}^6 + \frac{C}{K} \, \|\nabla_d \varepsilon(s)\|_{L^2}^2 + CKh^{-6} \, \|\varepsilon(s)\|_{L^{10}}^{10} \, .$$

Proof. Recall that

$$\begin{aligned}\|S(s)\|_{L^2}^2 &= \sum_{\sigma_{ij}} |\langle ij \rangle| \left| \frac{S_{ij}(s)}{d_{ij}} \right|^2 \\ &= \sum_{\sigma_{ij}} |\langle ij \rangle| \, \frac{|\sin(\theta_j(s) - \theta_i(s)) - (\theta_j(s) - \theta_i(s))|^2}{d_{ij}^4}.\end{aligned}$$

There exists c such that $|\sin(\varphi) - \varphi| \leq c\varphi^3$, and using that $\theta_i(s) = \theta(s, x_i) + \varepsilon_i$, we can bound for some C:

$$|\sin(\theta_j(s) - \theta_i(s)) - (\theta_j(s) - \theta_i(s))|^2 \leq C \, |\theta(s, x_j) - \theta(s, x_i)|^6 + C \, |\varepsilon_j(s) - \varepsilon_i(s)|^6 \, .$$

Plugging this in the sum yields

$$\begin{aligned}\|S(s)\|_{L^2}^2 &\leq C \sum_{\sigma_{ij}} |\langle ij \rangle| \, \frac{|\theta(s, x_j) - \theta(s, x_i)|^6}{d_{ij}^4} + C \sum_{\sigma_{ij}} |\langle ij \rangle| \, \frac{|\varepsilon_j(s) - \varepsilon_i(s)|^6}{d_{ij}^4} \\ &= I + II.\end{aligned}$$

Since $d_{ij} < h$, we have for the θ part:

$$\begin{aligned} I &\leq Ch^2 \sum_{\sigma_{ij}} |\langle ij \rangle| \left| \frac{\theta(s, x_j) - \theta(s, x_i)}{d_{ij}} \right|^6 \\ &\leq Ch^2 \sum_{\sigma_{ij}} |\langle ij \rangle| \, \|\nabla \theta(s)\|_{L^\infty}^6 \\ &\leq Ch^2 \, \|\nabla \theta(s)\|_{L^\infty}^6 \\ &\leq Ch^2 \, \|\theta(s)\|_{H^4}^6 \end{aligned}$$

because $\theta(s) \in H^4(\Omega)$, so the gradient $\nabla\theta(s) \in H^3(\Omega) \hookrightarrow C(\bar{\Omega})$.

For the ε part, we write:

$$\begin{aligned} \frac{|\varepsilon_j - \varepsilon_i|^6}{d_{ij}^4} &= \frac{|\varepsilon_j - \varepsilon_i|}{d_{ij}} \frac{|\varepsilon_j - \varepsilon_i|^5}{d_{ij}^3} \\ &\leq \frac{1}{2K} \frac{|\varepsilon_j - \varepsilon_i|^2}{d_{ij}^2} + \frac{1}{2} K \frac{|\varepsilon_j - \varepsilon_i|^{10}}{d_{ij}^6} \\ &\leq \frac{1}{2K} \frac{|\varepsilon_j - \varepsilon_i|^2}{d_{ij}^2} + \frac{1}{2} CK \frac{|\varepsilon_j|^{10} + |\varepsilon_i|^{10}}{d_{ij}^6} \end{aligned}$$

using Young's inequality. The second term can thus be bounded by:

$$\begin{aligned} II &\leq \frac{C}{K} \sum_{\sigma_{ij}} |\langle ij \rangle| \, \frac{|\varepsilon_j(s) - \varepsilon_i(s)|^2}{d_{ij}^2} + CK \sum_{\sigma_{ij}} |\langle ij \rangle| \, \frac{|\varepsilon_j(s)|^{10} + |\varepsilon_i(s)|^{10}}{d_{ij}^6} \\ &= \frac{C}{K} \, \|\nabla_d \varepsilon(s)\|_{L^2}^2 + CKh^{-6} \, \|\varepsilon(s)\|_{L^{10}}^{10} , \end{aligned}$$

by reorganizing the second sum, and using that $d_{ij} > \zeta h$. Combining estimates on I and II concludes. □

The previous lemma makes us want to apply a discrete Poincaré-Sobolev embedding to control the L^{10} norm. We will make use of the following result from [2]:

Lemma 4.6.4. *There exists C_{sob} only depending on Ω and ζ such that, for any $t \in [0, T]$:*

$$\|\varepsilon(t)\|_{L^{10}} \leq C_{sob} \left(\|\nabla_d \varepsilon(t)\|_{L^2} + \left| \int_\Omega \varepsilon(t, x) dx \right| \right)$$

Proof. Theorem 3 from [2] is the Poincaré-Sobolev embedding

$$\|\varepsilon(t)\|_{L^{10}} \leq C \left(\|\nabla_d \varepsilon(t)\|_{L^2} + \|\varepsilon(t)\|_{L^2} \right) .$$

We bound $\|\varepsilon(t)\|_{L^2}$ using the discrete Poincaré-Wirtinger inequality ([2], Theorem 5):

$$\left\| \varepsilon(t) - \overline{\varepsilon(t)} \right\|_{L^2} \leq C \, \|\nabla_d \varepsilon(t)\|_{L^2} ,$$

where $\overline{\varepsilon(t)} = \int_\Omega \varepsilon(t, x) dx$. □

Before continuing the proof, we also need to control the integral of $\varepsilon(s)$ over Ω in view of the previous inequality. This is done in the following lemma:

Lemma 4.6.5. *For any $t \in [0,T]$, the integral of $\varepsilon(t)$ satisfies for some C_I depending on θ, D and T:*

$$\left| \int_\Omega \varepsilon(t,x)dx \right| \leq C_I h.$$

Proof. Let $I(t) = \int_\Omega \varepsilon(t,x)dx$. We use the already derived equation for ε, that is:

$$\begin{aligned}\frac{d^2}{dt^2}\varepsilon_i(t) + D\frac{d}{dt}\varepsilon_i(t) = &\sum_{j\sim i} V_{ij}(\varepsilon_j(t) - \varepsilon_i(t)) \\ &+ \sum_{j\sim i} \frac{l_{ij}}{|P_i|}(S_{ij}(t) - R_{ij}(t)) + \sum_{j\sim i} G_{ij}\sin(\theta_j(t) - \theta_i(t)) - M_i(t).\end{aligned}$$

We multiply it by $|P_i|$ and sum over i to observe:

$$\begin{aligned}\frac{d^2}{dt^2}I(t) + D\frac{d}{dt}I(t) = &\sum_{i=1}^n |P_i| \sum_{j\sim i} V_{ij}(\varepsilon_j(t) - \varepsilon_i(t)) \\ &+ \sum_{i=1}^n |P_i| \sum_{j\sim i} l_{ij}(S_{ij}(t) - R_{ij}(t)) + \sum_{i=1}^n |P_i| \sum_{j\sim i} G_{ij}\sin(\theta_j(t) - \theta_i(t)) \\ &- \sum_{i=1}^n |P_i|\, M_i(t).\end{aligned}$$

By antisymmetry of the term $|P_i|\, V_{ij}(\varepsilon_j(t) - \varepsilon_i(t))$, the sum over all edges cancels out. The same happens to the second one, so only remains:

$$\frac{d^2}{dt^2}I(t) + D\frac{d}{dt}I(t) = \sum_{i=1}^n |P_i| \sum_{j\sim i} G_{ij}\sin(\theta_j(t) - \theta_i(t)) - \int_\Omega M(t,x)dx.$$

It is easy to see that we can bound each term:

$$\frac{d^2}{dt^2}I(t) + D\frac{d}{dt}I(t) \leq \|G\|_{L^2} + \|M(t)\|_{L^2}.$$

Since $I(0) = 0$ and

$$\frac{d^2}{dt^2}I(t) + D\frac{d}{dt}I(t) = e^{-Dt}\frac{d}{dt}\left(e^{Dt}\frac{d}{dt}I(t)\right),$$

we get a bound

$$e^{Dt}\frac{d}{dt}I(t) \leq C(\|G\|_{L^2} + \sup_{s\in[0,T]} \|M(s)\|_{L^2})$$

which in turn gives

$$I(t) \leq C(\|G\|_{L^2} + \sup_{s\in[0,T]} \|M(s)\|_{L^2}).$$

Using the hypothesis on G and Lemma 4.6.2 for M concludes. □

We can now finish the proof of Theorem 4.2.2 by combining all those inequalities.

4.7 Concluding the proof

We come back to the energy estimate on ε now that we have bounded every term. The next lemma ensures smallness of the discrete gradient through a bootstrap argument.

Lemma 4.7.1. *There exists h_0 depending on θ, C_G, Ω, D, T, δ and ζ, such that, if $h \leq h_0$, the following estimate holds for all $t \in [0,T]$:*

$$\|\nabla_d \varepsilon(t)\|_{L^2} \leq h^\delta.$$

Proof. Lemma 4.7.1 gives the bound

$$\begin{aligned}\|\nabla_d \varepsilon(t)\|_{L^2}^2 \leq C \int_0^t \|\partial_t R(s)\|_{L^2}^2 \, ds + C \|R(t)\|_{L^2}^2 \\ + C \int_0^t \|S(s)\|_{L^2}^2 \, ds + C \int_0^t \|M(s)\|_{L^2}^2 \, ds \\ + CT \|G\|_{L^2}^2 .\end{aligned}$$

Applying Lemma 4.6.1 and 4.6.2, as well as the hypothesis on G (5) immediately gives

$$\begin{aligned}\|\nabla_d \varepsilon(t)\|_{L^2}^2 \leq C \left(TC_R h^2 + C_R h^2 + C_M h^2 + TC_G h^2\right) \\ + C \int_0^t \|S(s)\|_{L^2}^2 \, ds.\end{aligned}$$

Now we can invoke Lemma 4.6.3, that is

$$\|S(s)\|_{L^2}^2 \leq Ch^2 \|\theta(s)\|_{H^4}^6 + \frac{C}{K} \|\nabla_d \varepsilon(s)\|_{L^2}^2 + CKh^{-6} \|\varepsilon(s)\|_{L^{10}}^{10},$$

Combining the Poincaré-Sobolev inequality (Lemma 4.6.4)

$$\|\varepsilon(s)\|_{L^{10}} \leq C_{sob} \left(\|\nabla_d \varepsilon(s)\|_{L^2} + \left| \int_\Omega \varepsilon(s,x) dx \right| \right),$$

with Lemma 4.6.5, that is:

$$\left| \int_\Omega \varepsilon(s,x) dx \right| \leq C_I h,$$

yields the bound

$$\|S(s)\|_{L^2}^2 \leq C \left(h^2 \|\theta(s)\|_{H^4}^6 + \frac{1}{K} \|\nabla_d \varepsilon(s)\|_{L^2}^2 + Kh^{-6} \|\nabla_d \varepsilon(s)\|_{L^2}^{10} + Kh^4 \right)$$

with a big constant C. With an even bigger constant C_θ, we have for any t:

$$\|\nabla_d \varepsilon(t)\|_{L^2}^2 \leq C_\theta \left(h^2 + \frac{1}{K} \sup_{s\in[0,t]} \|\nabla_d \varepsilon(s)\|_{L^2}^2 + Kh^{-6} \sup_{s\in[0,t]} \|\nabla_d \varepsilon(s)\|_{L^2}^{10} + Kh^4 \right)$$

Using that a supremum over $[0,t]$ is smaller than a supremum over $[0,\tau]$ if $t \leq \tau$, we can easily manipulate the previous inequality to obtain a supremum on the left hand side:

$$\sup_{s\in[0,t]} \|\nabla_d \varepsilon(s)\|_{L^2}^2 \leq C_\theta \left(h^2 + \frac{1}{K} \sup_{s\in[0,t]} \|\nabla_d \varepsilon(s)\|_{L^2}^2 + Kh^{-6} \sup_{s\in[0,t]} \|\nabla_d \varepsilon(s)\|_{L^2}^{10} + Kh^4 \right)$$

Now we choose $K = 2C_\theta$ so we can transfer to the left hand side the second term, and get, by bounding h^4 by h^2 and with a change of C_θ:

$$\sup_{s\in[0,t]} \|\nabla_d \varepsilon(s)\|_{L^2}^2 \leq C_\theta \left(h^2 + h^{-6} \sup_{s\in[0,t]} \|\nabla_d \varepsilon(s)\|_{L^2}^{10} \right).$$

Let $h_0 \leq 1$ be such that $C_\theta h_0^{2-2\delta} \leq \frac{1}{4}$ (notice that $2 - 2\delta > 0$). If $h \leq h_0$, we have:

$$\sup_{s\in[0,t]} \|\nabla_d \varepsilon(s)\|_{L^2}^2 \leq \frac{1}{4} \left(h^{2\delta} + h^{2\delta-8} \sup_{s\in[0,t]} \|\nabla_d \varepsilon(s)\|_{L^2}^{10} \right).$$

This gives the desired bound by contradiction: suppose there exists t_* such that

$$\|\nabla_d \varepsilon(t_*)\|_{L^2} > h^\delta.$$

At time $t = 0$, $\|\nabla_d \varepsilon(0)\|_{L^2} = 0$, and $\|\nabla_d \varepsilon(t)\|_{L^2}$ is a continuous function of time, so the explosion time

$$\tau = \inf\{\ t \mid \|\nabla_d \varepsilon(t)\|_{L^2} \geq h^\delta\}$$

is necessarily positive: $\tau > 0$. By definition, for any $0 \leq t < \tau$ we have $\|\nabla_d \varepsilon(t)\|_{L^2} \leq h^\delta$, and by continuity this holds for $t = \tau$. Thus

$$\sup_{s\in[0,\tau]} \|\nabla_d \varepsilon(s)\|_{L^2} \leq h^\delta.$$

Plugging this in the previous estimate,

$$\begin{aligned} \sup_{s\in[0,\tau]} \|\nabla_d \varepsilon(s)\|_{L^2}^2 &\leq \frac{1}{4} \left(h^{2\delta} + h^{2\delta-8} h^{10\delta} \right) \\ &= \frac{h^{2\delta}}{4} \left(1 + h^{10\delta-8} \right) \\ &\leq \frac{h^{2\delta}}{2} \end{aligned}$$

since by hypothesis $10\delta - 8 \geq 0$. But then, for all $s \in [0, \tau + \alpha]$ for α sufficiently small, $\|\nabla_d \varepsilon(t)\|_{L^2} < h^\delta$ by continuity. This contradicts the definition of τ. The lemma is proved. □

Remark 4.7.2. *The use of the Sobolev embedding $H^1 \hookrightarrow L^{10}$ (or any L^p with $p > 6$, the choice $p = 10$ is quite arbitrary) is the only reason that prevents us from extending the statements to dimension $d = 3$. In three-dimensional space, we only have up to $H^1 \hookrightarrow L^6$, which happens to be the endpoint at which the argument fails: the lower bounds and upper bounds needed on δ collapse, because the estimate in Lemma 4.6.3 then has the term $h^{-4} \|\varepsilon(s)\|_{L^6}^6$. Following the bootstrap argument, we would need $h^{-4+6\delta} \leq h^{2\delta+\epsilon}$ for some $\epsilon > 0$ to make the prefactor constant small. This would require $\delta > 1$ which is not compatible with the bound $h^2 \leq h^{2\delta+\epsilon}$ that we need for the other terms.*

We are now only a few inequalities away from Theorem 4.2.2:

Proof of Theorem 4.2.2. We pick h_0 given by Lemma 4.7.1. Using the discrete Poincaré-Wirtinger inequality ([2], Theorem 5):

$$\begin{aligned}\|\varepsilon(t)\|_{L^2} &\leq C\left(\|\nabla_d \varepsilon(t)\|_{L^2} + \left|\int_\Omega \varepsilon(t,x)dx\right|\right)\\ &\leq C\left(h^\delta + h\right)\\ &\leq Ch^\delta,\end{aligned}$$

where we used Lemmas 4.7.1 and 4.6.5 to control both quantities. Observe that, for $x \in P_i$,

$$|\theta(t,x) - \theta(t,x_i)| \leq \|\nabla\theta\|_{L^\infty} h,$$

so that

$$\|\theta(t) - \theta_d(t)\|_{L^2} \leq \|\nabla\theta\|_{L^\infty} h$$

where

$$\theta_d(t,x) = \sum_{i=1}^n \theta(t,x_i)\mathbf{1}_{P_i}(x).$$

Finally, we prove Theorem 4.2.2 by the triangular inequality:

$$\begin{aligned}\|\theta^n(t) - \theta(t)\|_{L^2} &\leq \|\varepsilon(t)\|_{L^2} + \|\theta_d(t) - \theta(t)\|_{L^2}\\ &\leq Ch^\delta\end{aligned}$$

where C depends on Ω, θ and ζ. □

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